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Mathematical Modeling of Anthrax Transmission Dynamics with Optimal Control: A Case of Anthrax in Human and Cattle Populations

Ebenezer Selorm Akplo (ebenezer.akplo@aims-cameroon.org)
African Institute for Mathematical Sciences (AIMS)
Cameroon

Supervised by: Dr. Stephen Lassong.
Kumasi Technical University, Ghana

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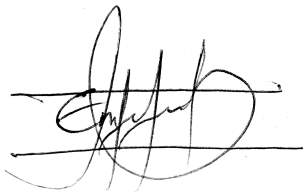
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Abstract

Anthrax, caused by *Bacillus anthracis*, remains a significant zoonotic threat to humans, domestic animals, and wild animals due to the bacterium's ability to form environmentally persistent spores and the role of carcasses in sustaining transmission. We develop a deterministic compartmental model that integrates livestock and human populations, environmental reservoirs, and carcass-mediated pathways to capture both direct and indirect routes of infection. The model incorporates carcass decomposition dynamics, enabling quantitative assessment of outbreak persistence and the relative contributions of transmission pathways. We establish fundamental model properties, including positivity and boundedness of solutions, derive the basic reproduction number $\mathcal{R}_0$, and characterize equilibrium states. Analytical results determine conditions for local and global stability of the disease-free equilibrium, and sensitivity analysis identifies key parameters driving transmission. An optimal control framework is formulated to evaluate intervention strategies such as vaccination, treatment, and carcass removal. Numerical results demonstrate that integrated measures especially combined vaccination, prompt treatment, and systematic carcass disposal are the most effective ways of reducing transmission, providing clear guidance for anthrax control policy.

Declaration

I, the undersigned, hereby declare that the work contained in this essay is my original work, and that any work done by others or by myself previously has been acknowledged and referenced accordingly.



Ebenezer Selorm Akplo, 23 June 2026.

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1. Introduction

1.1 Background of the study

Anthrax is a deadly zoonotic infectious disease caused by the bacterium *Bacillus anthracis* [5, 18]. This disease primarily affects herbivorous animals such as cattle, sheep, and goats [5]. However, animals such as dogs and cats are at risk of getting infected when they eat anthrax-contaminated animal products, but they normally do recover without treatment [5]. Anthrax can also be transmitted to humans through direct or indirect contact with infected animals or contaminated animal products [5, 18]. *Bacillus anthracis* exists in two main forms which are the vegetative form and a highly resistant spore form [5]. The highly resistant spores upon exposure to environmental conditions such as oxygen remain viable in soil for several decades [18]. The anthrax *Bacillus* has the ability to form spores with a thick outer coating which shields to protect itself when conditions are unfavorable [5, 18].

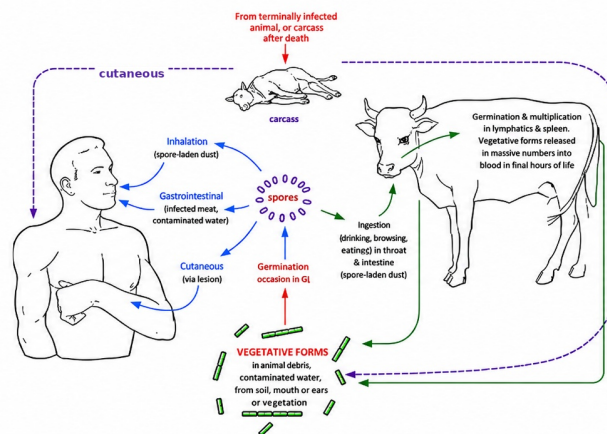


Figure 1.1: Schematic representation of the anthrax transmission cycle

Anthrax infection manifests in three main distinct ways. Cutaneous anthrax, the most prevalent form, develops when spores enter the skin through wounds or abrasions, and is typically characterized by a painless ulcer with a black center. Gastrointestinal anthrax infection occurs when one consumes contaminated animal products such as meat. This form of anthrax is associated with symptoms such as pain, vomiting, and diarrhea [10]. Inhalational anthrax is the most dangerous form, acquired through spore inhalation into the lungs, characterized by initial flu-like symptoms that progress to severe respiratory failure and frequently results in death without treatment [10, 18].

Although there are three main ways of contracting anthrax through infected animals, there are very few cases where anthrax infection has come from other sources. One notable example of such cases is the 2001 anthrax bioterrorism attack in the United States in which spores of powdered anthrax were intentionally spread throughout the U.S. postal service and resulted in 22 confirmed cases of anthrax infection with 5 deaths [10]. In addition to this example, there have also been reported outbreaks of anthrax through the drug injection route [18, 10], which illustrates the many possible means of contracting anthrax; however, for most people, the primary means of infection remains zoonotic and environmental].

The transmission dynamics of anthrax is very complex due to multiple pathways where it can be transmitted that is, from environmental reservoirs to animal host species, and the interaction between humans and animal hosts. In most cases of anthrax infection, the main route of transmission to livestock is through ingestion of contaminated soil, vegetation, or water. After coming into contact with these spores, infected animals can then transmit the disease to humans. Humans become infected via contact with carcasses of an infected animal, eating contaminated animal products or exposure to contaminated animal products such as hides [18]. Anthrax is different from other infectious diseases because anthrax transmission does not typically transmit from human to human, therefore the primary transmission routes for anthrax remain zoonotic and environmental pathways.

1.2 Problem statement.

Anthrax can spread from animals to humans since it is a zoonotic illness. The bacterium *Bacillus anthracis*, which can last in the environment for extended periods of time, is the cause. Anthrax is primarily spread through animal carcasses. Animal carcasses play a central role in the transmission of anthrax. When infected animals die, their bodies release spores into the surrounding environment during decomposition. Over time, these spores can accumulate in the soil, forming localized hotspots where grazing animals are at greater risk of infection. Humans may also contract the disease through contact with contaminated animals or animal products, allowing anthrax to persist within communities [4]. Despite the well-established importance of carcasses in the transmission cycle, many existing models either overlook them or treat them as part of the general environmental compartment. This approach fails to fully capture the unique role carcasses play in sustaining and amplifying the spread of the disease [5, 13, 14]. This approach fails to capture the pulsed release of spores during decomposition, a process that can significantly accelerate the spread of the disease. Studies have shown that sites with infected carcasses can remain infectious for long periods, increasing the risk of transmission, especially for grazing animals [16]. Anthrax primarily spreads through contact with a contaminated environment. When an infected animal dies and its carcass decomposes, spores are released into the soil and nearby vegetation. Humans can contract the infection by coming into close contact with contaminated carcasses or animal products, and grazing animals can consume these spores while feeding. This highlights the central role that carcasses play in sustaining the disease. Additionally, because anthrax spores can persist and accumulate in the environment for long periods, it is essential to account for carcasses when studying how the disease spreads [3]. Therefore, there is a need for a mathematical model that includes a carcass compartment in anthrax transmission models. This addition makes the model more biologically realistic, offers a more accurate depiction of environmental contamination, and strengthens its capacity to inform and evaluate effective anthrax control strategies.

1.3 Research questions

1. How can the dynamics of anthrax in human and cattle populations be studied using a mathematical model?
2. What are the equilibrium points of the proposed model?
3. What is the basic reproduction number $\mathcal{R}_0$ of the developed mathematical model?
4. Which model parameters influence anthrax transmission the most?

5. What control measures would be the most effective and how would they influence anthrax transmission dynamics?

Objectives of the Study.

General objective. To develop and analyze a mathematical model that explains the dynamics of anthrax transmission in humans and cattle populations.

Specific objectives.

1. To formulate a mathematical model for transmission dynamics of anthrax.
2. To calculate the equilibrium points and analyze their stability.
3. To compute the basic reproduction number (R_0) of the proposed model.
4. To carry out sensitivity analysis to determine how parameters affect the spread of anthrax.
5. To evaluate the effects of various control strategies.

Significance and scope of the study. This study contributes to the knowledge of transmission mechanisms of anthrax in both animals and humans populations. It addresses the factors controlling bacterial spread through mathematical modeling and then proposes control methods. This study is useful for health professionals and policymakers for the control, prevention and cure of anthrax and also aid in designing efficient vaccine campaigns and treatment measures for controlling the disease. The scope of this study is modeling the transmission dynamics between cattle and human populations considering the interrelationship between animal, human, and also environmental reservoir, and covered mathematical and numerical aspects. In summary, this study aims at providing useful information on the dynamics of anthrax transmission and contribute to further studies to help in reducing the burden of the disease in both animals and human.

1.4 Organization of the study

The rest of the work is organized into four chapters. Chapter 2 provides a review of related work on anthrax spread, mathematical modeling of infectious diseases, and also highlights the contributions of related research works. Chapter 3 presents the mathematical framework, theorems, and approaches utilized in the research. The theoretical and numerical analysis of the anthrax model for this study and findings from this study are presented in Chapter 4. Chapter 5 summarizes the findings, draws conclusions, and makes recommendations for further study.

2. Literature review

This chapter examines mathematical models of anthrax transmission that have been published, with an emphasis on models that use a deterministic compartmental framework to show how humans, animals, and the infected environment interact.

2.1 Review of related studies.

The highly resistant spores of *Bacillus anthracis* are a major factor in the spread of anthrax. These spores can remain viable in soil for up to several decades, waiting to be ingested or inhaled by a susceptible host [15, 18]. According to [18], anthrax transmission arises as a result of two main factors: changes in the environment and the season of the year. Changes in environment contribute to the growth and persistence of anthrax spores *Bacillus anthracis* in the environment due to soil factors such as soil pH, temperature, and the presence of soil minerals, such as calcium [18]. Changes in season and ecological factors, such as the availability of forage for grazing or the health status of the hosts, further affect the transmission of the disease [18].

In most cases, Anthrax infections arise from herbivorous mammals, but it can also be transmitted to humans, making anthrax a zoonotic disease [5, 13]. Animals become infected through ingestion, cutaneous transmission via skin, or inhalation when they are exposed to the resistant spores [15]. Once exposed to anthrax spores, the carcasses of the infected animal contribute significantly to environmental contamination and, under favorable conditions, continue as the primary source of anthrax spores. According to [15, 18] infections among susceptible hosts arise via direct contact with contaminated grazing areas, watering holes, or carcasses of infected animals when not properly disposed of. Humans become infected by direct contact with infected mammals or through the consumption of infected or contaminated mammal products (e.g., consumption of raw meat, contamination of hides, hair and/or skin). Reports by [10, 18] state that 95% of patients display cutaneous signs and symptoms, while 5% of patients exhibit signs and symptoms associated with inhalation or gastrointestinal transmission.

Mathematical models are essential in understanding how infectious diseases are transmitted among humans and animals. These models help us to construct the frameworks through which the spread of infectious diseases is represented mathematically [13]. In mathematical epidemiology, one of the most common approaches in infectious disease modeling is the use of compartmental models [1]. The foundation for compartmental mathematical modeling of infectious diseases begins with Kermack and McKendrick (1927) who introduced the classical Susceptible-Infected-Recovered (SIR) mathematical modelling framework [1]. According to compartmental representation, a population is defined as a system of interacting sub-groups; thus, each compartment corresponds to a specific epidemiological state. Individuals move between defined compartments through clearly defined transition pathways. These movements of a population between compartments illustrates the dynamics of the infectious disease's spread within that system [11].

For many compartmental modeling of infectious diseases, systems of ordinary differential equations (ODEs) are used to explain the transmission dynamics of disease. The deterministic modeling approach allows for the assumption of homogeneous mixing within a population and allows continuous-time representations of disease dynamics. ODE-based compartmental framework can be expanded in various ways. These may include delay differential equations (DDEs) to account for incubation and/or latent periods. Incorporating stochastic differential equations (SDEs) to model the randomness associated with

infectious disease spread, particularly among smaller populations or social groups. Partial differential equations (PDEs) to illustrate the effects of spatial heterogeneity. Additionally, fractional differential equations (FDEs) are incorporated to include memory effects, reflecting how current disease dynamics depend on past states of the system [11]. A key concept in the compartmental models is "the basic reproduction number (R_0)" [17], which refers to the expected number of secondary infections produced by one infected individual (I) in an entirely susceptible (S) population. This number is a dd use to determine whether an infectious disease will increase in prevalence or be eliminated from a population [17]. Overall, compartmental models provide a flexible and systematic foundation for the representation of infectious disease dynamics. They provide the grounds for us to incorporate biological realism by the addition of new compartments or modification of existing compartments to understand disease dynamics.

A model developed by [5] was a nonlinear deterministic mathematical model to study the transmission dynamics of anthrax among animal and human populations with an environmental compartment to capture the presence of an environmental reservoir contaminated with anthrax spores . The model developed in their study assumes that the path of transmission of the anthrax infectious agent (*Bacillus anthracis*) to an animal population was by way of contact with an environmental reservoir contaminated by anthrax spores. According to [5] when an animal is infected with anthrax, that animal becomes symptomatic of the disease and sheds the pathogen into the environment, contributing to the larger pool of available anthrax spores, which will subsequently infect other susceptible hosts. A major distinction of the model developed by [5] was the incorporation of a saturation incidence to model the crowding effect associated with the interaction of infected animals with one another and a factor in the model that adjusted the effect of animal contact behavior as the infectious disease progressed to its peak incidence in the population.

The results of the sensitivity analysis of the study of [5] indicated that to limit the transmission of anthrax, it is necessary that policies that promote increased hygiene and sanitation in the environment be established and implemented. These policy activities such as increased hygiene and sanitation contribute to increasing the decay rate of pathogens and minimizing the infectious force on susceptible animal populations. In addition, the saturation effect was applied to reduce the rate of infections in both human and animal populations; thus, inverse relationships were seen between behavioural changes and the population of infected humans. This shows that negative behavioural change increases the infected population of humans, while positive behavioural change decreases the infected human population over time. Similar findings are noted in terms of the saturation factor, where the level of pathogens in the environment decreases as the saturation factor increases.

However, the model developed in their study [5] does not provide for a carcass compartment; therefore, it does not take into account the role of infected animal carcasses as a means of transmission of the disease. This omission may limit the model's ability to accurately measure short-term dynamics of anthrax outbreaks because carcasses are a major source of environmental contamination and are critical to the persistence of the disease in endemic regions.

A Deterministic compartmental model was used by [13] to study the transmission dynamics of anthrax in both human and animal populations, and it included vaccination of susceptible animals within the vector compartments. The model used in their study assumes that anthrax transmission will occur through direct contact between infectious animals and humans as well as through the ingestion of contaminated animal products. One unique aspect of the work of [13] is the incorporation of vaccination with waning immunity in animals. Analyses of the model show that the disease-free equilibrium is stable when the basic reproduction number is less than one, but there is an endemic equilibrium when it is greater than one. The study indicated that backward bifurcation is possible, meaning that the disease can become established even if the reproduction number is below one. Hence, a conclusion regarding anthrax

dynamics based on only the reproduction number could be misleading.

Sensitivity analyses revealed that transmission rates and the recruitment rates of animals significantly affect the spread of disease while improved recovery rates and increased vaccination efforts are expected to decrease the burden of disease. The work of [13] provides a comparison of the rate of infection and the recovery rate of anthrax in humans and animals. The results indicate that the recovery rate of infection in animals is faster than the recovery rate of infection in humans. However, the model does not provide a carcass or reservoir for the environmental transmission of spores, which limits the ability of the model to capture short-term dynamics and environmental persistence of anthrax outbreak events.

[14] used a delayed mathematical model to investigate the dynamics of an anthrax epidemic. The model used delay differential equations (DDEs) to represent the incubation period of the disease as well. The delay parameters introduced in this research included a natural delay (the amount of time between when an animal or human becomes exposed and when they become infectious) and an artificial delay (the amount of time between when a public health intervention is instituted and when it has an effect on the disease). The results of their study show that both transmission parameters and the delay parameter/s affect the basic reproduction number, but that the values of the reproduction number decrease exponentially as the artificial delay parameter increases. Thus, results from this research provide a better understanding of disease transmission in animal populations and how intervention measures, such as quarantine, and restrictions on animal movement, can successfully reduce the transmission of the disease.

The sensitivity analyses conducted in their study [14] found that limiting contact within high-risk areas or between infected animals and uninfected animals is the most effective of the intervention strategies evaluated. Furthermore, the results from the numerical simulations demonstrated that the level of delay significantly contributes to reducing the infected population over time, and that higher levels of delay move the infected population toward zero. Thus, the results support the need for incorporating delay mechanisms into the understanding of disease transmission.

In spite of the contributions of the study of [14], there exist limitations in these existing models that limit their application. First, only animal populations are included in the model, so it cannot be used to address the zoonotic aspects of anthropomorphic transmission of anthrax.

2.2 Limitations of existing models and research gap

As highlighted in the review of related studies, existing models of anthrax transmission dynamics demonstrate both strengths and limitations. There are models that include animal and human populations, behaviors and environmental contamination, but the lack of an account for carcass-mediated transmission limits the potential of those models to capture short-term outbreaks due to infected carcasses. Also the work of [13] that incorporate vaccination and waning immunity provide critical insights into control efforts, but those models lack the environmental and carcass compartments that contribute to environmental reservoirs of anthrax. Also, [14] models provide a detailed view of delayed mechanism for the persistence and temporal dynamics of anthrax but there is no consideration of the human compartment, limiting the understanding of the zoonotic aspects of anthrax transmission.

In summary, a fundamental gap exists in the current literature regarding the dynamics of anthrax. Most of the studies conducted to now have investigated one or two of these factors in isolation rather than addressing them collectively. This study seeks to address this gap by providing a compartmental model that incorporates animal and human interactions, carcass transmission and environmental persistence.

3. Research Methodology

The mathematical framework and computational approach used in this study are presented in this chapter. The mathematical concepts and tools discussed in this chapter provide the foundation for the model development and analysis presented in Chapter 4 and also support the numerical results obtained.

3.1 Theoretical framework

Dynamical system. A dynamical system in mathematics is a rule that is used to explain how a system's state changes over time. Mathematically dynamic system is of the form:

$$\phi : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

Given that:

$E \in \mathbb{R}^n$ the state space and $\phi(t, x_0)$ is the system's state space with a starting time from an initial condition $x_0 \in E$.

The mapping ϕ is continuously differentiable C^1 and satisfies the following properties :

1. $\phi(0, x_0) = x_0, \quad \forall x_0 \in E$

We call it the identity function defined by $\phi(x_0) = x_0$. Which implies that at time $t = 0$, the system remains at its initial state.

2. $\phi(t + s, x) = \phi(t, \phi(s, x)), \quad \forall t, s \in \mathbb{R}, x \in E$

which indicates that evolving the system for time s and then for time t is equivalent to evolving it directly for time $t + s$.

The system is referred to as a continuous dynamical system when time t is described at continuous intervals, whereas it is referred to as a discrete dynamical system when time is treated as discrete (for instance, $t \in \mathbb{Z}$). In this study, a continuously dynamical system is employed since disease transmission occurs continuously.

The anthrax model's state space in this study is a state vector that includes all possible ways to divide the population into distinct compartments ($x(t) = (S_a, I_a, R_a, S_h, I_h, R_h, E, C)^T \in \mathbb{R}_+^8$).

The initial condition $x(0) = x_0 \geq 0$ in this study represents the sizes of all eight compartments at the start of the observation period. Understanding the concept of dynamical system framework is essential for analyzing the movement in and out of compartments over time.

Differential Equation. A differential equation is a mathematical relationship that expresses how dependent variables change in relation to an independent variable. Derivatives are used to express these rates of change.

Generally there are two categories of differential equations:

- Ordinary Differential Equations (ODEs): ODEs involve derivatives with regard to a single dependent variable, such as time (t).

- Partial Differential Equations (PDEs): PDEs include derivatives with regard to multiple independent variables, such as space and time.

In mathematical modeling, Ordinary differential equations are commonly used to describe how dependent variables changes in relation to other independent variables [11]. In this study ordinary differential equation are used to study the changes in the compartments over time.

3.1.1 Definition. Let Ω be an open set in the space $\mathbb{R}^n$, and let $(g : \Omega \rightarrow \mathbb{R}^n)$ be a continuously differentiable function. This function is called a vector field. The differential equation is then of the form:

$$x' = g(x), \quad x(0) = x_o \quad (3.1.1)$$

When the function (g) does not explicitly rely on an independent variable (eg time), the system is referred to as autonomous. If not, it is non-autonomous[9]. In this study autonomous differential equation is exclusively used.

equation (3.1.1), can also be written as:

$$\frac{dx_i}{dt} = g(x_1, x_2, \dots, x_n) \quad x_i(0) = x_{io} \quad i = 1, 2, \dots, n \quad (3.1.2)$$

The system is referred to as an Initial Value-Problem (IVP) if we have the initial conditions, that is, the system's value as the starting time. However, the system is referred to as a boundary value problem (BVP) if we have both the beginning and the ending conditions.

When differential equations are interrelated we call them a "system of differential equations". A system of differential equations. is a set of n interrelated differential equations [9]. This system with n compartments can be represented as:

$$\begin{cases} \frac{dx_1}{dt} = g_1(x_1, x_2, \dots, x_n) \\ \frac{dx_2}{dt} = g_2(x_1, x_2, \dots, x_n) \\ \vdots \\ \frac{dx_n}{dt} = g_n(x_1, x_2, \dots, x_n) \end{cases} \quad (3.1.3)$$

In our compartmental model, system (3.1.3) , takes $n = 8$, corresponding to the eight compartments $(S_a, I_a, R_a, S_h, I_h, R_h, E, C)$ to better help understand the dynamics of anthrax .

Compartmental Models. In mathematical modeling, models for transmission of disease are broadly classified as either stochastic or deterministic, depending on how randomness is incorporated in the study. This study employs, a deterministic modeling framework to examine the transmission dynamics of anthrax.

A deterministic model is a kind of mathematical model where the system's parameters and initial conditions completely dictate the result. Randomness does not exist in deterministic modeling [1]. Deterministic models of infectious diseases employ compartmental frameworks that separate the population into discrete epidemiological classes according to disease status.

In this study, the population is partitioned into eight compartments representing a distinct disease states. Susceptible animals, infected animals, and recovered animals. Susceptible humans, infected humans, recovered humans. Carcass and environment as a compartment to understand the dynamics of environmental transmission. This study uses a deterministic approach because this approach allows us to understand the movement between these compartments over time.

Equilibrium Points. Consider a system of ordinary differential equations given by:

$$\frac{dx}{dt} = f(x),$$

where $x \in \mathbb{R}^n$. A point x^* is called an equilibrium point if $f(x^*) = 0$. This corresponds to a constant solution of the system, meaning that the state of the system does not change over time. Equilibria are often among the most important of differential equation [9].

In the context of disease modeling, two types of equilibrium are important for understanding the dynamics of a disease:

1. A disease-free equilibrium (DFE). When there is no infection in the population, the system is said to be in a disease-free equilibrium. In this study, the disease-free equilibrium can also be called anthrax-free equilibrium. Setting all diseased compartments to zero yields the disease-free equilibrium [5].
2. A disease endemic equilibrium (DEE). An equilibrium point is said to be a disease endemic point if the disease persists in the population over time. At this point, one or more infected compartments remain positive [5].

Basic Reproduction Number $\mathcal{R}_0$. The basic reproduction number (R_0), is the average number of secondary infections caused by an infected person in a population of individuals who are fully susceptible. [17, 5].

$\mathcal{R}_0$ is used in epidemic modeling to calculate how fast a disease could spread in its early stages, when everyone is at risk. Using $\mathcal{R}_0$:

- If $\mathcal{R}_0 < 1$, each infected individual on average creates less than one new infection over the period of being infected.
- If $\mathcal{R}_0 = 1$, each infected person produces one new infection throughout infectious period. The disease prevalence in the population tends to remain at a constant level.
- If $\mathcal{R}_0 > 1$, each infected person, on average produces more than one new infection. As a result, the disease may spread throughout the populace and perhaps lead to an epidemic.

Computation of the Basic Reproduction Number. The basic reproduction number R_0 can be computed using; Jacobian method and "Next Generation Matrix method".

In this study, the Next Generation Matrix approach is employed due to its systematic structure and its wide applicability in compartmental epidemic models. The Next Generation Matrix method involves splitting the system of equations into two components: the rate of appearance of new infections and the transfer of individuals between compartments.

Using the Next Generation Matrix approach we have the matrices F and V defined as:

$$F = \left[\frac{\partial f_i}{\partial x_j} \right], \quad V = \left[\frac{\partial v_i}{\partial x_j} \right]$$

where f_i represents the rate of new infections in compartment i , and v_i represents the transfer of individuals into and out of compartment i .

Using this approach, the basic reproduction number is given by

$$R_0 = \rho(FV^{-1})$$

where $\rho(FV^{-1})$ denotes the spectral radius (dominant eigenvalue) of the matrix FV^{-1} . The basic reproduction number is given by the dominant eigenvalue of the matrix FV^{-1} .

3.2 Stability theory

In this work, we examine two forms of stability; the local stability and the global stability. While the global stability provides information about the system's behavior throughout the entire domain, the local stability provides information about the system's behavior close to the equilibrium point.

Local Stability of the Disease-Free Equilibrium. Using local stability analysis, we can investigate the system's behavior close to an equilibrium point. Local stability of the disease-free equilibrium (DFE) for disease models indicates whether the disease will die out if there are initially very few infections. If the DFE is locally stable when $\mathcal{R}_0 < 1$, small changes in infection levels will not cause the disease to take hold.

To investigate the local asymptotic stability of the disease-free equilibrium $\mathcal{E}_0$ of the anthrax model, this study employ the next generation method of [17]. This method uses the Jacobian matrix evaluated at the DFE to determine stability conditions.

Global Stability of the Disease-Free Equilibrium. Having established that the local asymptotic stability guarantees that the disease-free equilibrium (DFE) is stable to small incidence fluctuations when $\mathcal{R}_0 < 1$, this condition does not rule out the possibility that large incidence fluctuations could lead to the disease not dying out even when $\mathcal{R}_0 < 1$. To establish a stronger result, the study investigates the *global* asymptotic stability of the DFE. Global stability ensures that the disease dies out *regardless of the initial conditions*, provided $\mathcal{R}_0 < 1$.

To achieve this, the method developed by [8] was used to examine the global asymptotic stability of the DFE. The method by [8] requires that the disease transmission model (in this case anthrax model) be written in the following form:

$$\begin{aligned} \frac{d\mathbf{x}}{dt} &= F(\mathbf{x}, \mathbf{I}), \\ \frac{d\mathbf{I}}{dt} &= G(\mathbf{x}, \mathbf{I}), \quad G(\mathbf{x}, \mathbf{0}) = 0, \end{aligned} \tag{3.2.1}$$

where $\mathbf{x} \in \mathbb{R}^m$ denotes the vector of uninfected compartments and $\mathbf{I} \in \mathbb{R}^n$ denotes the vector of infected compartments (including latent, infectious, and environmental pathogen classes) [8]. The condition $G(\mathbf{x}, \mathbf{0}) = 0$ ensures that the subspace of the disease-free is invariant; that is, if no infection is present, no new infection can arise spontaneously. $\mathbf{U}_0 = (\mathbf{x}^*, \mathbf{0})$ denotes the disease-free equilibrium point of the system [8].

3.3 Numerical simulation and analysis

There are several numerical methods for solving ordinary differential equations (ODEs), including Taylor's Method, Euler's Method and Runge-Kutta's Method. In this study the Runge-Kutta 4th/5th-order (RK45) method via SciPy's `solve_ivp` was used. This study used the Runge-Kutta method because its order of approximation to a solution is better compared to other forms of numerical methods such as Euler's method. The Runge-Kutta method is used to solve ordinary differential equations (ODEs) when initial conditions are given. This study used initial conditions and made simulations over a 500 day period.

After simulation of baseline (without control), the study then proceeds to a sensitivity analysis.

Sensitivity Analysis. Sensitivity analysis identifies which parameters most influence the basic reproduction number $\mathcal{R}_0$. This helps us to know which control measures are the most effective in reducing epidemic risk. For this study we use the normalized sensitivity index of $\mathcal{R}_0$ with respect to a parameter p is defined as

$$\Upsilon_p^{\mathcal{R}_0} = \frac{\partial \mathcal{R}_0}{\partial p} \cdot \frac{p}{\mathcal{R}_0} \quad \text{where } p \text{ is a model parameter}$$

The normalized forward sensitivity (NFS) index quantifies the relative impact of each parameter on $\mathcal{R}_0$. A positive value of $\Upsilon_p^{\mathcal{R}_0}$ means the parameter p increases $\mathcal{R}_0$ while a negative value means the parameter p decreases $\mathcal{R}_0$.

The sensitivity indices for all parameters in $\mathcal{R}_0$ were then presented visually in a bar chart generated from the numerical code. This chart helps us to visualize which parameters have the strongest influence on the basic reproduction number ($\mathcal{R}_0$). Parameters observed to exceed the threshold line shown by red dashed lines are highly sensitive. This sensitivity result was a guide on the choice of optimal control strategies to be implemented.

4. Model, analysis, results and discussion

This chapter presents the formulated model, the analysis of the equilibrium points, the reproduction number R_0 , study of the stability and sensitivity of the model and numerical simulation.

4.1 Model formulation and description

Through the transmission dynamics of anthrax, the total population was split into eight compartments: animals who are at risk of being infected ($S_a(t)$), animals who are already infected ($I_a(t)$), animals who have recovered from the disease ($R_a(t)$), carcasses of dead animals ($C(t)$), the environment ($E(t)$), susceptible humans ($S_h(t)$), humans who are infected ($I_h(t)$), and recovered humans ($R_h(t)$),

The model for this study was developed based on the framework established by [5]. The study specifically adopted their approach to modeling the environmental reservoir of anthrax pathogens. The study extended the model of [5] by introducing a Carcass (C) compartment. The Carcass compartment (C) alongside the environment compartment (E), allows for a much better understanding of the transmission dynamics of anthrax. These two compartments help in differentiating between the short-term and long-term dynamics of the disease.

Model Assumptions. The assumptions underlying this model are:

- ($H1$) All animals and humans are susceptible to the disease. The population has no prior immunity.
- ($H2$) Recruitment of new animals and humans occurs through birth and migration at constant rates.
- ($H3$) There is no incubation period. Infected individuals become immediately infectious upon pathogen exposure, allowing direct transition from susceptible to infectious compartments.
- ($H4$) Transmission occurs only from animals to humans or through environmental routes; human-to-human transmission does not occur.
- ($H5$) Humans and animals experience natural mortality independent of disease status at rates μ_h and μ_a respectively.
- ($H6$) Immunity acquired after recovery is temporary; recovered individuals lose immunity and return to susceptibility at rates π_a and π_h respectively.
- ($H7$) Infected mothers do not transmit infection to offspring; vertical transmission is absent in both populations.

The anthrax transmission model for this study is shown in figure 4.1

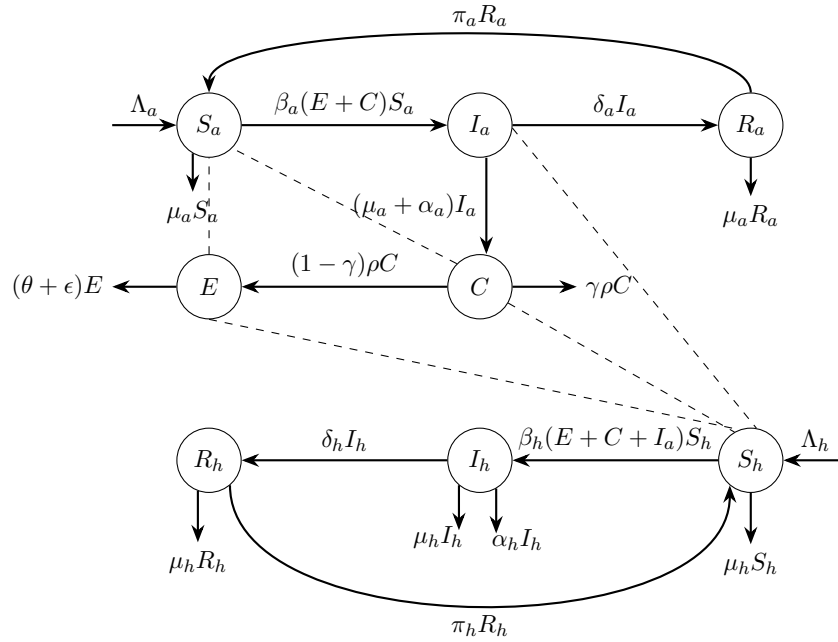


Figure 4.1: Anthrax transmission dynamics.

Parameter Description.

Parameter	Description
Λ_a	Recruitment rate of susceptible animals
μ_a	Natural death rate of animals
β_a	Transmission rate for animals via contact with E and C
δ_a	Recovery rate of infected animals
α_a	Disease-induced death rate of infected animals
π_a	Rate of waning immunity in animals
Λ_h	Recruitment rate of susceptible humans
μ_h	Natural death rate of humans
β_h	Transmission rate for humans via contact with E , C , and I_a
δ_h	Recovery rate of infected humans
α_h	Disease-induced death rate of infected humans
π_h	Rate of waning immunity in humans
ρ	Overall carcass removal rate
γ	Proportion of carcasses safely removed ($0 \leq \gamma \leq 1$)
ϵ	Natural spore decay rate in the environment
θ	Spore removal rate due to decontamination

Table 4.1: Description of model parameters

Transmission Dynamics

The anthrax model for this study integrates animal and human populations through environmental contamination and carcasses. Susceptible animals are recruited at rate Λ_a and experience natural death at rate μ_a . Susceptible animals are infection via contact with environmental spores (E) and carcasses (C) at transmission rate β_a . Animals upon infection either recover at rate δ_a or die from anthrax at rate α_a . Recovered animals lose immunity at rate π_a , returning to the susceptible class. Carcasses contribute to environmental contamination: unsafe carcasses (proportion $1 - \gamma$) remain as infection sources, while safely removed carcasses (proportion γ) are eliminated at rate ρ . Humans are recruited at rate Λ_h and experience natural death at rate μ_h . They acquire anthrax infection through three pathways: contact with environmental spores (E), contaminated carcasses (C), and infected animals (I_a), all at transmission rate β_h . Infected humans either recover at rate δ_h or die from anthrax at rate α_h . Recovered humans then lose immunity at rate π_h . Environmental spores decay naturally at rate ϵ or are removed through decontamination at rate θ .

Model Equation. from Figure 4.1, we obtain the differential equation system as:

$$\begin{cases} \frac{dS_a}{dt} = \Lambda_a - \beta_a(E + C)S_a - \mu_a S_a + \pi_a R_a \\ \frac{dI_a}{dt} = \beta_a(E + C)S_a - (\delta_a + \alpha_a + \mu_a)I_a \\ \frac{dR_a}{dt} = \delta_a I_a - (\mu_a + \pi_a)R_a \\ \frac{dS_h}{dt} = \Lambda_h - \beta_h(E + C + I_a)S_h - \mu_h S_h + \pi_h R_h \\ \frac{dI_h}{dt} = \beta_h(E + C + I_a)S_h - (\delta_h + \mu_h + \alpha_h)I_h \\ \frac{dR_h}{dt} = \delta_h I_h - (\mu_h + \pi_h)R_h \\ \frac{dE}{dt} = (1 - \gamma)\rho C - (\theta + \epsilon)E \\ \frac{dC}{dt} = (\mu_a + \alpha_a)I_a - \rho C \end{cases} \quad (4.1.1)$$

Positivity and Boundedness. For an epidemic model to be useful, it must represent the real-life dynamics of the disease. Since we are dealing with a population, the number of people or animals in each compartment must remain non-negative at all times.

Positivity of solutions. We begin by showing that all solutions of the model remain positive.

4.1.1 Theorem. *Let the initial conditions of the system Figure 4.1 be $\{S_{a0} \geq 0, I_{a0} \geq 0, R_{a0} \geq 0, S_{h0} \geq 0, I_{h0} \geq 0, R_{h0} \geq 0, E_0 \geq 0, C_0 \geq 0\}$. Then the set $\{S_a(t), I_a(t), R_a(t), S_h(t), I_h(t), R_h(t), E(t), C(t)\}$ of equation (4.1.1) is positive for all $t > 0$.*

Proof. Taking into consideration the model equation (4.1.1), we then prove the positivity of the state variables applying the procedure of [12].

Susceptible Animal Compartment. We consider the compartment of susceptible animals.

$$\frac{dS_a}{dt} = \Lambda_a - \beta_a(E(t) + C(t))S_a - \mu_a S_a + \pi_a R_a.$$

Since $\Lambda_a \geq 0$ and $\pi_a R_a \geq 0$,

$$\begin{aligned}
&\implies \dot{S}_a \geq -(\beta_a(E(t) + C(t)) + \mu_a)S_a(t), \\
&\implies \frac{\dot{S}_a(t)}{S_a(t)} \geq -(\beta_a(E(t) + C(t)) + \mu_a), \quad S_a(t) \neq 0, \\
&\implies \int \frac{\dot{S}_a(t)}{S_a(t)} dt \geq - \int (\beta_a(E(t) + C(t)) + \mu_a) dt, \\
&\implies \ln S_a(t) \geq - \int (\beta_a(E(t) + C(t)) + \mu_a) dt + \ln K, \\
&\implies S_a(t) \geq K \cdot e^{-\int (\beta_a(E(t) + C(t)) + \mu_a) dt}, \\
&\implies S_a(t) \geq S_a(0) \cdot e^{-\int (\beta_a(E(t) + C(t)) + \mu_a) dt} > 0.
\end{aligned}$$

Evaluating at $t = 0$, we have $K = S_a(0) > 0$, thus:

$$S_a(t) \geq S_a(0) e^{-\int_0^t (\beta_a(E(t) + C(t)) + \mu_a) dt}$$

Since $S_a(0) > 0$ and the exponential term is always positive, we conclude $S_a(t) > 0$ for all $t \geq 0$.

Using Integration factor method we then proof the positivity of the carcass compartment. This is given by:

$$\begin{aligned}
&\frac{dC}{dt} = (\mu_a + \alpha_a)I_a - \rho C, \\
&\text{Since } (\mu_a + \alpha_a)I_a \geq 0 \implies \frac{dC}{dt} \geq -\rho C(t), \\
&\implies \frac{dC}{dt} + \rho C(t) \geq 0.
\end{aligned}$$

Let $e^{\rho t}$ be an integration factor. Multiply both sides by $e^{\rho t}$:

$$\begin{aligned}
&\implies e^{\rho t} \frac{dC}{dt} + \rho C(t) e^{\rho t} \geq 0, \\
&\implies \frac{d}{dt} (C(t) e^{\rho t}) \geq 0, \\
&\implies \int_0^t \frac{d}{ds} (C(s) e^{\rho s}) ds \geq 0, \\
&\implies [C(s) e^{\rho s}]_0^t \geq 0, \\
&\implies C(t) e^{\rho t} - C(0) e^0 \geq 0, \\
&\implies C(t) e^{\rho t} \geq C(0).
\end{aligned}$$

Therefore:

$$C(t) \geq C(0) e^{-\rho t} \geq 0.$$

□

Since $C(0) \geq 0$ and the exponential term is always positive, $C(t) \geq 0$ for all $t \geq 0$.

Therefore $S_a(t)$ stays positive for $t \geq 0$. Using the same procedure for other state variables we have :

$$I_a(t) \geq I_a(0) e^{-(\delta_a + \alpha_a + \mu_a)t} \geq 0, R_a(t) \geq R_a(0) e^{-(\mu_a + \pi_a)t} \geq 0, I_h(t) \geq I_h(0) e^{-(\delta_h + \mu_h + \alpha_h)t} \geq 0, R_h(t) \geq R_h(0) e^{-(\mu_h + \pi_h)t} \geq 0, E(t) \geq E(0) e^{-(\theta + \varepsilon)t} \geq 0.$$

The detailed proofs for I_a, R_a, S_h, I_h, R_h , and E are provided in Appendix (A).

Boundedness. For the model to be biologically meaningful, its solutions must remain bounded and non-negative for all time. We therefore establish the boundedness of the solutions.

4.1.2 Theorem. Let $\Omega = \{S_a(t), I_a(t), R_a(t), S_h(t), I_h(t), R_h(t), E(t), C(t) \in \mathbb{R}_+^8 : S_a(0) \geq 0, I_a(0) \geq 0, R_a(0) \geq 0, S_h(0) \geq 0, I_h(0) \geq 0, R_h(0) \geq 0, E(0) \geq 0, C(0) \geq 0, N_a(t) \leq \frac{\Lambda_a}{\mu_a}, N_h(t) \leq \frac{\Lambda_h}{\mu_h}\}$ is positively invariant and attracting the solutions to the system of equation (4.1.1).

Proof. We consider the animal Population □

$$\begin{aligned} N_a(t) &= S_a(t) + I_a(t) + R_a(t) \\ \frac{dN_a(t)}{dt} &= \frac{dS_a(t)}{dt} + \frac{dI_a(t)}{dt} + \frac{dR_a(t)}{dt} \end{aligned}$$

$$\implies \frac{dN_a}{dt} = \Lambda_a - \beta_a(E + C)S_a - \mu_a S_a + \pi_a R_a + \beta_a(E + C)S_a - (\delta_a + \alpha_a)I_a + \delta_a I_a - (\mu_a + \pi_a)R_a$$

$$\implies \frac{dN_a}{dt} = \Lambda_a - \mu_a(S_a + I_a + R_a) - \alpha_a I_a$$

$$\implies \frac{dN_a}{dt} = \Lambda_a - \mu_a(N_a(t)) - \alpha_a I_a$$

$$\implies \frac{dN_a}{dt} \leq \Lambda_a - \mu_a(N_a(t)) \quad \text{since } I_a \geq 0$$

$$\implies \frac{dN_a}{dt} + \mu_a(N_a(t)) \leq \Lambda_a$$

let $e^{\mu_a t}$ be an integration factor. and then we multiply through by the integration factor.

$$\begin{aligned}
&\Rightarrow e^{\mu_a t} \frac{dN_a}{dt} + \mu_a (N_a(t)) e^{\mu_a t} \leq \Lambda_a e^{\mu_a t} \\
&\Rightarrow \int_0^t \frac{d}{ds} (N_a(s) e^{\mu_a s}) ds \leq \int_0^t \Lambda_a e^{\mu_a s} ds \\
&\Rightarrow [N_a(s) e^{\mu_a s}]_0^t \leq \left[\Lambda_a \frac{1}{\mu_a} e^{\mu_a s} \right]_0^t \\
&\Rightarrow N_a(t) e^{\mu_a t} - N_a(0) e^{\mu_a \cdot 0} \leq \frac{\Lambda_a}{\mu_a} [e^{\mu_a t} - e^{\mu_a \cdot 0}] \\
&\Rightarrow N_a(t) e^{\mu_a t} - N_a(0) \leq \frac{\Lambda_a}{\mu_a} [e^{\mu_a t} - 1] \\
&\Rightarrow N_a(t) \leq \frac{\Lambda_a}{\mu_a} - \frac{\Lambda_a}{\mu_a} e^{-\mu_a t} + N_a(0) e^{-\mu_a t}
\end{aligned}$$

$N_a(t) \geq 0$ because for all $t \geq 0$ $S_a(t), I_a(t), R_a(t) \geq 0$ and also

$$N_a(t) = S_a(t) + I_a(t) + R_a(t)$$

$$\Rightarrow 0 \leq N_a(t) \leq \frac{\Lambda_a}{\mu_a} + e^{-\mu_a t} \left[N_a(0) - \frac{\Lambda_a}{\mu_a} \right]$$

As $t \rightarrow +\infty$, we have:

$$\begin{aligned}
0 \leq \lim_{t \rightarrow \infty} N_a(t) &\leq \lim_{t \rightarrow \infty} \left(\frac{\Lambda_a}{\mu_a} + e^{-\mu_a t} \left[N_a(0) - \frac{\Lambda_a}{\mu_a} \right] \right) \\
&\Rightarrow 0 \leq N_a(t) \leq \frac{\Lambda_a}{\mu_a} \quad \forall t \geq 0
\end{aligned}$$

Applying the same method to the human population (full computation is provided in Appendix A), we obtain:

$$\begin{aligned}
0 \leq \lim_{t \rightarrow \infty} N_h(t) &\leq \lim_{t \rightarrow \infty} \left(\frac{\Lambda_h}{\mu_h} + e^{-\mu_h t} \left[N_h(0) - \frac{\Lambda_h}{\mu_h} \right] \right) \\
&\Rightarrow 0 \leq N_h(t) \leq \frac{\Lambda_h}{\mu_h} \quad \forall t \geq 0
\end{aligned}$$

Therefore all solutions of system (4.1.1) starting in Ω are uniformly bounded and remain in the feasible region:

$$\Omega = \left\{ (S_a, I_a, R_a, S_h, I_h, R_h, E, C) \in \mathbb{R}_+^8 : N_a \leq \frac{\Lambda_a}{\mu_a}, N_h \leq \frac{\Lambda_h}{\mu_h} \right\} \quad (4.1.2)$$

4.2 Equilibrium analysis

Disease free equilibrium (DFE). When there is no diseases the dynamical system in (4.1.1) reach its Disease Free Equilibrium (DFE). We solve the system of equations in (4.1.1) by setting it to zero and then compute the DFE. Recovery and infections do not exist at DFE. Thus, $I_a = R_a = I_h = R_h = E = C = 0$. Based on these conditions we have a system :

$$\begin{cases} \Lambda_a - \mu_a S_a &= 0 \\ \Lambda_h - \mu_h S_h &= 0 \end{cases} \quad (4.2.1)$$

Solving the system of equations(4.2.1) we have:

$$\begin{aligned} 0 &= \Lambda_a - \mu_a S_a^* \\ \implies S_a^* &= \frac{\Lambda_a}{\mu_a} \end{aligned}$$

$$\begin{aligned} 0 &= \Lambda_h - \mu_h S_h^* \\ \implies S_h^* &= \frac{\Lambda_h}{\mu_h} \end{aligned}$$

$$\text{The DFE is : } \mathcal{E}_0 = \left[\frac{\Lambda_a}{\mu_a}, 0, 0, \frac{\Lambda_h}{\mu_h}, 0, 0, 0, 0 \right]^T$$

Disease endemic equilibrium (DEE). The point at which anthrax persists in a given population is known as the disease endemic equilibrium. It is derived by solving for the state variables after setting the derivatives in the model system (4.1.1) to zero. By setting the derivatives of (4.1.1) to zero we have:

$$\begin{cases} \Lambda_a - \beta_a(E + C)S_a - \mu_a S_a + \pi_a R_a &= 0 \\ \beta_a(E + C)S_a - (\delta_a + \alpha_a + \mu_a)I_a &= 0 \\ \delta_a I_a - (\mu_a + \pi_a)R_a &= 0 \\ \Lambda_h - \beta_h(E + C + I_a)S_h - \mu_h S_h + \pi_h R_h &= 0 \\ \beta_h(E + C + I_a)S_h - (\delta_h + \mu_h + \alpha_h)I_h &= 0 \\ \delta_h I_h - (\mu_h + \pi_h)R_h &= 0 \\ (1 - \gamma)\rho C - (\theta + \epsilon)E &= 0 \\ (\mu_a + \alpha_a)I_a - \rho C &= 0 \end{cases} \quad (4.2.2)$$

Solving the system of equations (4.2.2) :

We consider the carcass compartment

$$\begin{aligned} (\mu_a + \alpha_a)I_a^* - \rho C^* &= 0, \\ \implies C^* &= \frac{(\mu_a + \alpha_a)I_a^*}{\rho}. \end{aligned}$$

For the environment compartment we have:

$$\begin{aligned}
 (1 - \gamma)\rho C^* - (\theta + \epsilon)E^* &= 0, \\
 E^* &= \frac{(1 - \gamma)\rho}{(\theta + \epsilon)} C^*, \\
 \text{but: } C^* &= \frac{(\mu_a + \alpha_a)}{\rho} I_a^*, \\
 \Rightarrow E^* &= \frac{(1 - \gamma)\rho}{(\theta + \epsilon)} \times \frac{(\mu_a + \alpha_a)}{\rho} I_a^*, \\
 E^* &= \frac{(1 - \gamma)(\mu_a + \alpha_a)}{(\theta + \epsilon)} I_a^*.
 \end{aligned}$$

Using similar method we have the other endemic equilibrium as:

$$\left\{ \begin{aligned}
 S_a^* &= \frac{(\delta_a + \alpha_a + \mu_a)}{\beta_a(\mu_a + \alpha_a) \left(\frac{1}{\rho} + \frac{1-\gamma}{\theta+\epsilon} \right)}, \\
 I_a^* &= \frac{m_2 \Lambda_a - \mu_a m_1}{m_2 \left(m_1 - \frac{\delta_a \pi_a}{\mu_a + \pi_a} \right)}, \\
 R_a^* &= \frac{\delta_a}{(\mu_a + \pi_a)} I_a^*, \\
 S_h^* &= \frac{m_3 I_h^*}{\beta_h [(\mu_a + \alpha_a) m_4 + 1] I_a^*}, \\
 R_h^* &= \frac{\alpha_h}{\mu_h + \pi_h} I_h^*, \\
 I_h^* &= \frac{\Lambda_h}{k_3 - \frac{\pi_h \alpha_h}{\mu_h + \pi_h} + \frac{\mu_h m_3}{\beta_h [(\mu_a + \alpha_a) m_4 + 1] I_a^*}}
 \end{aligned} \right.$$

where $m_1 = (\delta_a + \alpha_a + \mu_a)$, $m_2 = \beta_a(\mu_a + \alpha_a) \left(\frac{1}{\rho} + \frac{1-\gamma}{\theta+\epsilon} \right)$, $m_3 = \delta_h + \mu_h + \alpha_h$ and $m_4 = \frac{1}{\rho} + \frac{1-\gamma}{\theta+\epsilon}$

The EE is : $[S_a^*, I_a^*, R_a^*, S_h^*, I_h^*, R_h^*, E^*, C^*]^T$

Basic Reproduction Number. We discussed the Next Generation Matrix approach for calculating R_0 in Chapter 3. Only the compartments directly related to the disease's spread are included in this computation [17]. Taking into account just the infectious compartments in the differential equation system in (4.1.1) we have:

$$\left\{ \begin{aligned}
 \frac{dI_a}{dt} &= \beta_a(E + C)S_a - (\delta_a + \alpha_a + \mu_a)I_a \\
 \frac{dI_h}{dt} &= \beta_h(E + C + I_a)S_h - (\delta_h + \mu_h + \alpha_h)I_h \\
 \frac{dE}{dt} &= (1 - \gamma)\rho C - (\theta + \epsilon)E \\
 \frac{dC}{dt} &= (\mu_a + \alpha_a)I_a - \rho C
 \end{aligned} \right. \quad (4.2.3)$$

Let $\mathcal{F}$ and $\mathcal{V}$ denote the number of new infections that enter and exit the system, respectively. From the system of equations (4.2.3) we have:

$$\mathcal{F} = \begin{pmatrix} \beta_a(E+C)S_a \\ \beta_h(E+C+I_a)S_h \\ 0 \\ 0 \end{pmatrix}, \quad \mathcal{V} = \begin{pmatrix} (\delta_a + \mu_a + \alpha_a)I_a \\ (\delta_h + \mu_h + \alpha_h)I_h \\ -(1-\gamma)\rho C + (\theta + \epsilon)E \\ -(\mu_a + \alpha_a)I_a + \rho C \end{pmatrix}$$

To find F and V , we recall that at the DFE (4.2.1) $(S_a^*, I_a^*, R_a^*, S_h^*, I_h^*, R_h^*, E^*, C^*) = (\frac{\Lambda_a}{\mu_a}, 0, 0, \frac{\Lambda_h}{\mu_h}, 0, 0, 0, 0)$

$$F = \frac{\partial \mathcal{F}}{\partial (I_a, I_h, E, C)} \Big|_{DFE} = \begin{pmatrix} 0 & 0 & \frac{\beta_a \Lambda_a}{\mu_h} & \frac{\beta_a \Lambda_a}{\mu_h} \\ \frac{\beta_h \Lambda_h}{\mu_h} & 0 & \frac{\beta_h \Lambda_h}{\mu_h} & \frac{\beta_h \Lambda_h}{\mu_h} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$V = \frac{\partial \mathcal{V}}{\partial (I_a, I_h, E, C)} \Big|_{DFE} = \begin{pmatrix} \delta_a + \mu_a + \alpha_a & 0 & 0 & 0 \\ 0 & \delta_h + \mu_h + \alpha_h & 0 & 0 \\ 0 & 0 & \theta + \epsilon & -(1-\gamma)\rho \\ -(\mu_a + \alpha_a) & 0 & 0 & \rho \end{pmatrix}$$

Computing for V^{-1} , for simplicity let:

$$k_1 = \delta_a + \alpha_a + \mu_a, \quad k_2 = \delta_h + \mu_h + \alpha_h, \quad k_3 = \theta + \epsilon, \quad k_4 = (1-\gamma)\rho, \quad k_5 = \mu_a + \alpha_a$$

$$\text{Hence } V = \begin{pmatrix} k_1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 \\ 0 & 0 & k_3 & -k_4 \\ -k_5 & 0 & 0 & \rho \end{pmatrix}$$

Using Gaussian elimination. We then use the Gaussian elimination method to solve the inverse of V

$$\left[\begin{array}{cccc|cccc} k_1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & k_3 & -k_4 & 0 & 0 & 1 & 0 \\ -k_5 & 0 & 0 & \rho & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{k_5 R_1 + k_1 R_4 \rightarrow R_4} \left[\begin{array}{cccc|cccc} k_1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & k_3 & -k_4 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & k_1 \rho & k_5 & 0 & 0 & k_1 \end{array} \right]$$

$$\left[\begin{array}{cccc|cccc} k_1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & k_3 & -k_4 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & k_1 \rho & k_5 & 0 & 0 & k_1 \end{array} \right] \xrightarrow{\frac{1}{k_1} R_1 \rightarrow R_1, \frac{1}{k_2} R_2 \rightarrow R_2} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{k_2} & 0 & 0 \\ 0 & 0 & k_3 & -k_4 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & k_1 \rho & k_5 & 0 & 0 & k_1 \end{array} \right]$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{k_2} & 0 & 0 \\ 0 & 0 & k_3 & -k_4 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & k_1 \rho & k_5 & 0 & 0 & k_1 \end{array} \right] \xrightarrow{k_1 \rho R_3 + k_4 R_4 \rightarrow R_3} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{k_2} & 0 & 0 \\ 0 & 0 & k_1 k_3 \rho & 0 & k_5 k_4 & 0 & k_1 \rho & k_1 k_4 \\ 0 & 0 & 0 & k_1 \rho & k_5 & 0 & 0 & k_1 \end{array} \right]$$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{k_2} & 0 & 0 \\ 0 & 0 & k_1 k_3 \rho & 0 & k_5 k_4 & 0 & k_1 \rho & k_1 k_4 \\ 0 & 0 & 0 & k_1 \rho & k_5 & 0 & 0 & k_1 \end{array} \right] \xrightarrow{\frac{1}{k_1 k_3 \rho} R_3 \mapsto R_3, \frac{1}{k_1 \rho} R_4 \mapsto R_4} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{k_2} & 0 & 0 \\ 0 & 0 & 1 & 0 & \frac{k_5 k_4}{k_1 k_3 \rho} & 0 & \frac{1}{k_3} & \frac{k_4}{k_3 \rho} \\ 0 & 0 & 0 & 1 & \frac{k_5}{k_1 \rho} & 0 & 0 & \frac{1}{\rho} \end{array} \right]$$

Therefore:

$$V^{-1} = \begin{pmatrix} \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & \frac{1}{k_2} & 0 & 0 \\ \frac{k_5 k_4}{k_1 k_3 \rho} & 0 & \frac{1}{k_3} & \frac{k_4}{k_3 \rho} \\ \frac{k_5}{k_1 \rho} & 0 & 0 & \frac{1}{\rho} \end{pmatrix} \quad \text{but since: } k_4 = (1 - \gamma)\rho \implies V^{-1} = \begin{pmatrix} \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & \frac{1}{k_2} & 0 & 0 \\ \frac{k_5 k_4}{k_1 k_3} & 0 & \frac{1}{k_3} & \frac{k_4}{k_3} \\ \frac{k_5}{k_1 \rho} & 0 & 0 & \frac{1}{\rho} \end{pmatrix}$$

$$\implies FV^{-1} = \begin{pmatrix} 0 & 0 & \frac{\beta_a \Lambda_a}{\mu_a} & \frac{\beta_a \Lambda_a}{\mu_a} \\ \frac{\beta_h \Lambda_h}{\mu_h} & 0 & \frac{\beta_h \Lambda_h}{\mu_h} & \frac{\beta_h \Lambda_h}{\mu_h} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{k_1} & 0 & 0 & 0 \\ 0 & \frac{1}{k_2} & 0 & 0 \\ \frac{k_5 k_4}{k_1 k_3} & 0 & \frac{1}{k_3} & \frac{k_4}{k_3} \\ \frac{k_5}{k_1 \rho} & 0 & 0 & \frac{1}{\rho} \end{pmatrix}$$

$$\implies FV^{-1} = \begin{pmatrix} \frac{\beta_a \Lambda_a}{\mu_a} \left(\frac{k_5 k_4}{k_1 k_3} + \frac{k_5}{k_1 \rho} \right) & 0 & \frac{\beta_a \Lambda_a}{\mu_a} \left(\frac{1}{k_3} \right) & \frac{\beta_a \Lambda_a}{\mu_a} \left(\frac{k_4}{k_3} + \frac{1}{\rho} \right) \\ \frac{\beta_h \Lambda_h}{\mu_h} \left(\frac{1}{k_1} + \frac{k_5 k_4}{k_1 k_3} + \frac{k_5}{k_1 \rho} \right) & 0 & \frac{\beta_h \Lambda_h}{\mu_h} \left(\frac{1}{k_3} \right) & \frac{\beta_h \Lambda_h}{\mu_h} \left(\frac{k_4}{k_3} + \frac{1}{\rho} \right) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The basic reproduction number is the spectral radius of FV^{-1} , $\implies \mathcal{R}_0 = \rho(FV^{-1})$:

$$\det(FV^{-1} - \Lambda I_4) = \begin{pmatrix} \frac{\beta_a \Lambda_a}{\mu_a} \left(\frac{k_5 k_4}{k_1 k_3} + \frac{k_5}{k_1 \rho} \right) - \Lambda_1 & 0 & \frac{\beta_a \Lambda_a}{\mu_a} \left(\frac{1}{k_3} \right) & \frac{\beta_a \Lambda_a}{\mu_a} \left(\frac{k_4}{k_3} + \frac{1}{\rho} \right) \\ \frac{\beta_h \Lambda_h}{\mu_h} \left(\frac{1}{k_1} + \frac{k_5 k_4}{k_1 k_3} + \frac{k_5}{k_1 \rho} \right) & -\Lambda_2 & \frac{\beta_h \Lambda_h}{\mu_h} \left(\frac{1}{k_3} \right) & \frac{\beta_h \Lambda_h}{\mu_h} \left(\frac{k_4}{k_3} + \frac{1}{\rho} \right) \\ 0 & 0 & -\Lambda_3 & 0 \\ 0 & 0 & 0 & -\Lambda_4 \end{pmatrix}$$

$$\implies \det(FV^{-1} - \Lambda I_4) = (-\Lambda_4)(-\Lambda_3) \begin{pmatrix} \frac{\beta_a \Lambda_a}{\mu_a} \left(\frac{k_5 k_4}{k_1 k_3} + \frac{k_5}{k_1 \rho} \right) - \Lambda_1 & 0 \\ \frac{\beta_h \Lambda_h}{\mu_h} \left(\frac{1}{k_1} + \frac{k_5 k_4}{k_1 k_3} + \frac{k_5}{k_1 \rho} \right) & -\Lambda_2 \end{pmatrix}$$

$$\Lambda_2 = \Lambda_3 = \Lambda_4 = 0 \quad \Lambda_1 = \frac{\beta_a \Lambda_a}{\mu_a} \left(\frac{k_5 k_4}{k_1 k_3} + \frac{k_5}{k_1 \rho} \right)$$

$$\implies \mathcal{R}_0 = \frac{\beta_a \Lambda_a (\mu_a + \alpha_a)}{\mu_a (\delta_a + \alpha_a + \mu_a)} \left(\frac{1 - \gamma}{\theta + \epsilon} + \frac{1}{\rho} \right) \quad (4.2.4)$$

4.2.1 Remark. The basic reproduction number $\mathcal{R}_0$ is the average number of secondary infections generated by a single infectious animal in a completely susceptible population. It is the key threshold determining whether anthrax will die out or persist.

Disease spread within the animal population through contact with environmental contamination and carcasses. The proportion of unsafe carcasses $((1 - \gamma))$ relative to spore removal rates. Higher unsafe carcass proportions and lower removal rates increase environmental contamination contribution to $\mathcal{R}_0$. The average duration contaminated carcasses remain infectious before removal is $\frac{1}{\rho}$. Longer persistence times increase the disease transmission potential.

4.3 Stability analysis

Local stability at DFE. This study uses the next generation approach of [17] to examine the local asymptotic stability of the disease-free equilibrium of system (4.1.1). According to [17], we arrange the compartments so that the first m compartments corresponds to infected individuals:

$$\mathbf{x} = (I_a, I_h, E, C, S_a, R_a, S_h, R_h)^T.$$

$$J(\mathcal{E}_0) = \begin{pmatrix} -x_1 & 0 & \beta_a S_a^0 & \beta_a S_a^0 & 0 & 0 & 0 & 0 \\ \beta_h S_h^0 & -x_3 & \beta_h S_h^0 & \beta_h S_h^0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -x_5 & (1 - \gamma)\rho & 0 & 0 & 0 & 0 \\ \mu_a + \alpha_a & 0 & 0 & -\rho & 0 & 0 & 0 & 0 \\ 0 & 0 & -\beta_a S_a^0 & -\beta_a S_a^0 & -\mu_a & \pi_a & 0 & 0 \\ \delta_a & 0 & 0 & 0 & 0 & -x_2 & 0 & 0 \\ -\beta_h S_h^0 & 0 & -\beta_h S_h^0 & -\beta_h S_h^0 & 0 & 0 & -\mu_h & \pi_h \\ 0 & \delta_h & 0 & 0 & 0 & 0 & 0 & -x_4 \end{pmatrix} \quad (4.3.1)$$

where : $x_1 = \delta_a + \alpha_a + \mu_a$, $x_2 = \mu_a + \pi_a$, $x_3 = \delta_h + \mu_h + \alpha_h$, $x_4 = \mu_h + \pi_h$, $x_5 = \theta + \epsilon$.

$$S_a^0 = \frac{\Lambda_a}{\mu_a}, \quad S_h^0 = \frac{\Lambda_h}{\mu_h}.$$

4.3.1 Theorem. The DFE, $\mathcal{E}_0$, is locally asymptotically stable if all eigenvalues of the matrix (4.3.1) have negative real parts and unstable if any eigenvalue of (4.3.1) has positive real part.

By Lemma 1 of [17], the eigenvalues of (4.3.1) can be partitioned into :

$$F - V = \begin{pmatrix} -x_1 & 0 & \beta_a S_a^0 & \beta_a S_a^0 \\ \beta_h S_h^0 & -x_3 & \beta_h S_h^0 & \beta_h S_h^0 \\ 0 & 0 & -x_5 & (1 - \gamma)\rho \\ \mu_a + \alpha_a & 0 & 0 & -\rho \end{pmatrix} \quad \text{and} \quad -J_4 = \begin{pmatrix} -\mu_a & \pi_a & 0 & 0 \\ 0 & -x_2 & 0 & 0 \\ 0 & 0 & -\mu_h & \pi_h \\ 0 & 0 & 0 & -x_4 \end{pmatrix} \quad (4.3.2)$$

where : $F - V$ corresponds to the infected compartments and $-J_4$ corresponds to the uninfected compartments. Also by Lemma 1 of [17], all the eigenvalues of $-J_4$ have negative real parts.

Proof. Since matrix $-J_4$ of (4.3.2) is an upper triangular, we then have its eigenvalues as :

$$\lambda_1 = -\mu_a < 0, \quad \lambda_2 = -x_2 < 0, \quad \lambda_3 = -\mu_h < 0, \quad \lambda_4 = -x_4 < 0.$$

hence $-J_4$ satisfies the condition of Lemma 1 of [17]

We then compute the eigenvalues of matrix $F - V$. The eigenvalues of $F - V$ is defined as:

$$P_I(\lambda) = \det(F - V - \lambda I_4) = 0.$$

$$\begin{aligned} \implies P_I(\lambda) &= (-\rho - \lambda) \begin{pmatrix} -x_1 - \lambda & 0 & \beta_a S_a^0 \\ \beta_h S_h^0 & -x_3 & \beta_h S_h^0 \\ 0 & 0 & -x_5 - \lambda \end{pmatrix} - ((\mu_a + \alpha_a) - \lambda) \begin{pmatrix} 0 & \beta_a S_a^0 & \beta_a S_a^0 \\ -x_3 & \beta_h S_h^0 & \beta_h S_h^0 \\ 0 & -x_5 & (1 - \gamma)\rho \end{pmatrix} \\ \implies P_I(\lambda) &= (\rho - \lambda)(-x_5 - \lambda) \begin{pmatrix} -x_1 - \lambda & 0 \\ \beta_h S_h^0 & -x_3 \end{pmatrix} - ((\mu_a + \alpha_a - \lambda)(-x_3 - \lambda)) \begin{pmatrix} \beta_a S_a^0 & \beta_a S_a^0 \\ -x_5 - \lambda & (1 - \gamma)\rho \end{pmatrix} \\ \implies P_I(\lambda) &= (-\rho - \lambda)(-x_5 - \lambda)(x_1 x_3 - \lambda) - ((\mu_a + \alpha_a - \lambda)(-x_3 - \lambda)) \beta_a S_a^0 ((1 - \gamma)\rho + x_5 + \lambda) \end{aligned}$$

We evaluate $P_I(\lambda)$ at $\lambda = 0$ to identify the stability threshold.

$$P_I(0) = \rho x_1 x_3 x_5 - (\mu_a + \alpha_a) x_3 \beta_a S_a^0 [(1 - \gamma)\rho + k_5]$$

$$\implies P_I(0) = \rho x_1 x_3 x_5 \left[1 - \frac{(\mu_a + \alpha_a) \beta_a S_a^0}{x_1} \left(\frac{1 - \gamma}{x_5} + \frac{1}{\rho} \right) \right] \quad (4.3.3)$$

The term in brackets is related to the basic reproduction number:

$$\mathcal{R}_0 = \frac{(\mu_a + \alpha_a) \beta_a S_a^*}{x_1} \left(\frac{1 - \gamma}{x_5} + \frac{1}{\rho} \right).$$

Therefore:

$$P_I(0) = \rho x_1 x_3 x_5 (1 - \mathcal{R}_0). \quad (4.3.4)$$

□

4.3.2 Remark. From equation (4.3.4), since $\rho, x_1, x_3, x_5 > 0$, by Theorem 2 of [17], the disease-free equilibrium $\mathcal{E}_0$ is locally asymptotically stable when $\mathcal{R}_0 < 1$ and unstable when $\mathcal{R}_0 > 1$. This establishes that the condition $\mathcal{R}_0 < 1$ is both necessary and sufficient for disease elimination near the disease-free equilibrium.

Global stability at DFE. As established in Chapter 3, the method of [8] requires rewriting the transmission model (4.1.1) in the form:

$$\begin{aligned} \frac{d\mathbf{x}}{dt} &= F(\mathbf{x}, \mathbf{I}), \\ \frac{d\mathbf{I}}{dt} &= G(\mathbf{x}, \mathbf{I}), \quad G(\mathbf{x}, \mathbf{0}) = 0, \end{aligned} \quad (4.3.5)$$

where $\mathbf{x} \in \mathbb{R}^4$ is $\mathbf{x} = (S_a, R_a, S_h, R_h)^T$ the vector of uninfected compartments and $\mathbf{I} \in \mathbb{R}^4$ is $\mathbf{I} = (I_a, I_h, E, C)^T$ the vector of infected compartments.

The DFE is $\mathbf{U}_0 = (\mathbf{x}^*, \mathbf{0})$ where $\mathbf{x}^* = \left(\frac{\Lambda_a}{\mu_a}, 0, \frac{\Lambda_h}{\mu_h}, 0 \right)$.

Writing the model equations (4.1.1), in the form of (4.3.5), we will have it as :

$$\begin{cases} \frac{dS_a}{dt} = \Lambda_a - \beta_a(E + C)S_a - \mu_a S_a + \pi_a R_a \\ \frac{dR_a}{dt} = \delta_a I_a - (\mu_a + \pi_a) R_a \\ \frac{dS_h}{dt} = \Lambda_h - \beta_h(E + C + I_a)S_h - \mu_h S_h + \pi_h R_h \\ \frac{dR_h}{dt} = \delta_h I_h - (\mu_h + \pi_h) R_h \end{cases} \quad (4.3.6)$$

and

$$\begin{cases} \frac{dI_a}{dt} = \beta_a(E + C)S_a - (\delta_a + \alpha_a + \mu_a)I_a \\ \frac{dI_h}{dt} = \beta_h(E + C + I_a)S_h - (\delta_h + \mu_h + \alpha_h)I_h \\ \frac{dE}{dt} = (1 - \gamma)\rho C - (\theta + \epsilon)E \\ \frac{dC}{dt} = (\mu_a + \alpha_a)I_a - \rho C \end{cases} \quad (4.3.7)$$

from (4.3.6) and (4.3.7) we have

$$F(\mathbf{x}, \mathbf{I}) = \begin{pmatrix} \Lambda_a - \beta_a(E + C)S_a - \mu_a S_a + \pi_a R_a \\ \delta_a I_a - (\mu_a + \pi_a) R_a \\ \Lambda_h - \beta_h(E + C + I_a)S_h - \mu_h S_h + \pi_h R_h \\ \delta_h I_h - (\mu_h + \pi_h) R_h \end{pmatrix} \quad \text{and}$$

$$G(\mathbf{x}, \mathbf{I}) = \begin{pmatrix} \beta_a(E + C)S_a - (\delta_a + \alpha_a + \mu_a)I_a \\ \beta_h(E + C + I_a)S_h - (\delta_h + \mu_h + \alpha_h)I_h \\ (1 - \gamma)\rho C - (\theta + \epsilon)E \\ (\mu_a + \alpha_a)I_a - \rho C \end{pmatrix}$$

from $G(\mathbf{x}, \mathbf{I})$, $G(\mathbf{x}, \mathbf{0}) = \mathbf{0}$: when $\mathbf{I} = (0, 0, 0, 0)^T$, confirming the condition in (4.3.5).

According to [8] Conditions (H1) and (H2) must be met to guarantee global asymptotic stability .

(H1) For $\frac{d\mathbf{x}}{dt} = F(\mathbf{x}, \mathbf{0})$, the equilibrium $\mathbf{x}^*$ is globally asymptotically stable (g.a.s.).

(H2) $G(\mathbf{x}, \mathbf{I}) = A\mathbf{I} - \hat{G}(\mathbf{x}, \mathbf{I})$, $\hat{G}(\mathbf{x}, \mathbf{I}) \geq 0$ for $(\mathbf{x}, \mathbf{I}) \in \Omega$,

where $A = D_{\mathbf{I}}G(\mathbf{x}^*, \mathbf{0})$ is an M-matrix with non-negative off-diagonal elements, and Ω is the feasible region of the model.

4.3.3 Theorem. *The fixed point $\mathbf{U}_0 = (\mathbf{x}^*, \mathbf{0})$ is a globally asymptotically stable (g.a.s.) equilibrium of system (4.3.5) provided that $\mathcal{R}_0 < 1$ (l.a.s.) and that Conditions (H1) and (H2) are satisfied:*

Verification of Condition (H1). When $\mathbf{I} = (0, 0, 0, 0)^T$, the uninfected system (4.3.6) reduces to:

$$\begin{cases} \frac{dS_a}{dt} = \Lambda_a - \mu_a S_a + \pi_a R_a \\ \frac{dR_a}{dt} = -(\mu_a + \pi_a) R_a \\ \frac{dS_h}{dt} = \Lambda_h - \mu_h S_h + \pi_h R_h \\ \frac{dR_h}{dt} = -(\mu_h + \pi_h) R_h \end{cases} \quad (4.3.8)$$

The system (4.3.8) consist of two independent subsystems: one for animals (S_a, R_a) and one for humans (S_h, R_h) .

Animal subsystem. From the R_a equation, we have:

$$\begin{aligned} \dot{R}_a(t) &= -(\mu_a + \pi_a) R_a \\ \implies \frac{\dot{R}_a(t)}{R_a} &= -(\mu_a + \pi_a) \\ \implies \int_0^t \frac{\dot{R}_a(s)}{R_a} ds &= \int_0^t -(\mu_a + \pi_a) ds \\ \implies [\ln(R_a(s))]_0^t &= [-(\mu_a + \pi_a)(s)]_0^t \\ \implies R_a(t) &= R_a(0) e^{-(\mu_a + \pi_a)t}, \text{ As } t \rightarrow \infty R_a(t) = 0 \end{aligned}$$

Substituting $R_a = 0$ into the $S_a(t)$ equation gives :

$$\begin{aligned} \frac{dS_a}{dt} &= \Lambda_a - \mu_a S_a \\ \implies \frac{dS_a}{dt} + \mu_a S_a &= \Lambda_a \end{aligned}$$

let $e^{\mu_a t}$ be an integration factor

$$\begin{aligned} \frac{dS_a}{dt} e^{\mu_a t} + \mu_a S_a e^{\mu_a t} &= \Lambda_a e^{\mu_a t} \\ \implies \frac{d}{dt} (S_a(t) e^{\mu_a t}) &= \Lambda_a e^{\mu_a t} \\ \int_0^t \frac{d}{ds} (S_a(s) e^{\mu_a s}) ds &= \Lambda_a \int_0^t e^{\mu_a s} ds \\ \implies S_a(t) e^{\mu_a t} - S_a(0) &= \frac{\Lambda_a}{\mu_a} e^{\mu_a t} - \frac{\Lambda_a}{\mu_a} \\ \implies S_a(t) &= \frac{\Lambda_a}{\mu_a} + \left[S_a(0) - \frac{\Lambda_a}{\mu_a} \right] e^{-\mu_a t} \end{aligned}$$

$$\text{As } t \rightarrow \infty, \quad S_a(t) = \frac{\Lambda_a}{\mu_a}$$

Human subsystem. Applying same method to the human subsystem, we obtain:

$$R_h(t) \longrightarrow 0, \quad R_h(t) = 0 \quad \text{and} \quad S_h(t) \longrightarrow \frac{\Lambda_h}{\mu_h} \quad \text{as } t \rightarrow \infty.$$

Therefore, $\mathbf{x}^* = \left(\frac{\Lambda_a}{\mu_a}, 0, \frac{\Lambda_h}{\mu_h}, 0 \right)$ is globally asymptotically stable, hence (H1) holds.

Verification of Condition (H2). We then continue to verify condition H2.

$$A = \begin{pmatrix} -(\delta_a + \alpha_a + \mu_a) & 0 & \frac{\beta_a \Lambda_a}{\mu_a} & \frac{\beta_a \Lambda_a}{\mu_a} \\ \frac{\beta_h \Lambda_h}{\mu_h} & -(\delta_h + \mu_h + \alpha_h) & \frac{\beta_h \Lambda_h}{\mu_h} & \frac{\beta_h \Lambda_h}{\mu_h} \\ 0 & 0 & -(\theta + \epsilon) & (1 - \gamma)\rho \\ (\mu_a + \alpha_a) & 0 & 0 & -\rho \end{pmatrix}$$

The off-diagonal components of A are nonnegative since: $\frac{\beta_a \Lambda_a}{\mu_a} \geq 0$, $\frac{\beta_h \Lambda_h}{\mu_h} \geq 0$, $(1 - \gamma)\rho \geq 0$, $(\mu_a + \alpha_a) \geq 0$ given that all model parameters are nonnegative. Hence A is an M-matrix as required.

$$A\mathbf{I} = \begin{pmatrix} -(\delta_a + \alpha_a + \mu_a)I_a + \frac{\beta_a \Lambda_a}{\mu_a}(E + C) \\ \frac{\beta_h \Lambda_h}{\mu_h}I_a - (\delta_h + \mu_h + \alpha_h)I_h + \frac{\beta_h \Lambda_h}{\mu_h}(E + C) \\ -(\theta + \epsilon)E + (1 - \gamma)\rho C \\ (\mu_a + \alpha_a)I_a - \rho C \end{pmatrix}$$

$$A\mathbf{I} - G(\mathbf{x}, \mathbf{I}) = \begin{pmatrix} -(\delta_a + \alpha_a + \mu_a)I_a + \frac{\beta_a \Lambda_a}{\mu_a}(E + C) \\ \frac{\beta_h \Lambda_h}{\mu_h}I_a - (\delta_h + \mu_h + \alpha_h)I_h + \frac{\beta_h \Lambda_h}{\mu_h}(E + C) \\ -(\theta + \epsilon)E + (1 - \gamma)\rho C \\ (\mu_a + \alpha_a)I_a - \rho C \end{pmatrix} - \begin{pmatrix} \beta_a(E + C)S_a - (\delta_a + \alpha_a + \mu_a)I_a \\ \beta_h(E + C + I_a)S_h - (\delta_h + \mu_h + \alpha_h)I_h \\ (1 - \gamma)\rho C - (\theta + \epsilon)E \\ (\mu_a + \alpha_a)I_a - \rho C \end{pmatrix}$$

$$\text{For } A\mathbf{I} - G(\mathbf{x}, \mathbf{I}) \quad \text{we have : } \hat{G}_1 = \frac{\beta_a \Lambda_a}{\mu_a}(E + C) - \beta_a(E + C)S_a = \beta_a(E + C) \left(\frac{\Lambda_a}{\mu_a} - S_a \right)$$

by similar computation:

$$\hat{G}(\mathbf{x}, \mathbf{I}) = \begin{pmatrix} \beta_a(E + C) \left(\frac{\Lambda_a}{\mu_a} - S_a \right) \\ \beta_h(E + C + I_a) \left(\frac{\Lambda_h}{\mu_h} - S_h \right) \\ 0 \\ 0 \end{pmatrix} \implies \hat{G}(\mathbf{x}, \mathbf{I}) = \begin{pmatrix} \beta_a(E + C)(S_a^* - S_a) \\ \beta_h(E + C + I_a)(S_h^* - S_h) \\ 0 \\ 0 \end{pmatrix} \quad (4.3.9)$$

because at the disease-free equilibrium $S_a^* = \frac{\Lambda_a}{\mu_a}$ and $S_h^* = \frac{\Lambda_h}{\mu_h}$, we can write:

$\widehat{G}(\mathbf{x}, \mathbf{I}) \geq 0$ In the biologically feasible region Ω (4.1.2), the susceptible populations satisfy:

$$S_a \leq S_a^* = \frac{\Lambda_a}{\mu_a} \quad \text{and} \quad S_h \leq S_h^* = \frac{\Lambda_h}{\mu_h},$$

$\widehat{G}_1 = \beta_a(E + C)(S_a^* - S_a) \geq 0$, $\widehat{G}_2 = \beta_h(E + C + I_a)(S_h^* - S_h) \geq 0$, $\widehat{G}_3 = \widehat{G}_4 = 0 \geq 0$, Hence $\widehat{G}(\mathbf{x}, \mathbf{I}) \geq 0$ for all $(\mathbf{x}, \mathbf{I}) \in \Omega$, hence condition **(H2)** is satisfied.

4.3.4 Remark. Therefore, since both conditions **(H1)** and **(H2)** holds, when $\mathcal{R}_0 < 1$, the disease-free equilibrium is globally asymptotically stable regardless of the initial conditions.

4.4 Sensitivity analysis and numerical simulation

Sensitivity analysis. We then perform sensitivity analysis to know how sensitive each parameter is in relation to $\mathcal{R}_0$. For sensitivity analysis and Numerical Simulation table 4.2 present the values used in this study.

Parameter	Value	Dimension	Source
Λ_a	0.035	day ⁻¹	Assumed
β_a	0.001-0.009	day ⁻¹	[2]
μ_a	0.002	day ⁻¹	Assumed
π_a	0.002	day ⁻¹	[6]
δ_a	0.0025	day ⁻¹	[2, 5, 6]
α_a	0.050	day ⁻¹	Assumed
Λ_h	0.021367	day ⁻¹	Assumed
β_h	0.0001	day ⁻¹	[2, 5, 6]
μ_h	0.000042734	day ⁻¹	[2]
π_h	0.001	day ⁻¹	Assumed
δ_h	0.04	day ⁻¹	[5]
α_h	0.0167	day ⁻¹	Assumed
ρ	0.03	day ⁻¹	Assumed
γ	0.5	dimensionless	Assumed
θ	0.045	day ⁻¹	[6, 7]
ϵ	0.8	day ⁻¹	[5]

Table 4.2: Parameter Values and Dimensions

Sensitivity Analysis. We then compute the sensitivity indices for all model parameters appearing in $\mathcal{R}_0$ (4.2.4) to determine the relative effect of each parameter on disease transmission.

A positive sensitivity index means the parameter increases $\mathcal{R}_0$ while a negative sensitivity index means the parameter, decreases $\mathcal{R}_0$.

Parameter	Sensitivity Index	Parameter	Sensitivity Index
Λ_a	1.000000	α_a	0.040401
β_a	1.000000	γ	-0.001772
μ_a	-0.907493	ε	-0.001533
ρ	-0.907480	θ	-0.000094
δ_a	-0.045662		

Table 4.3: Sensitivity Indices for Model Parameters

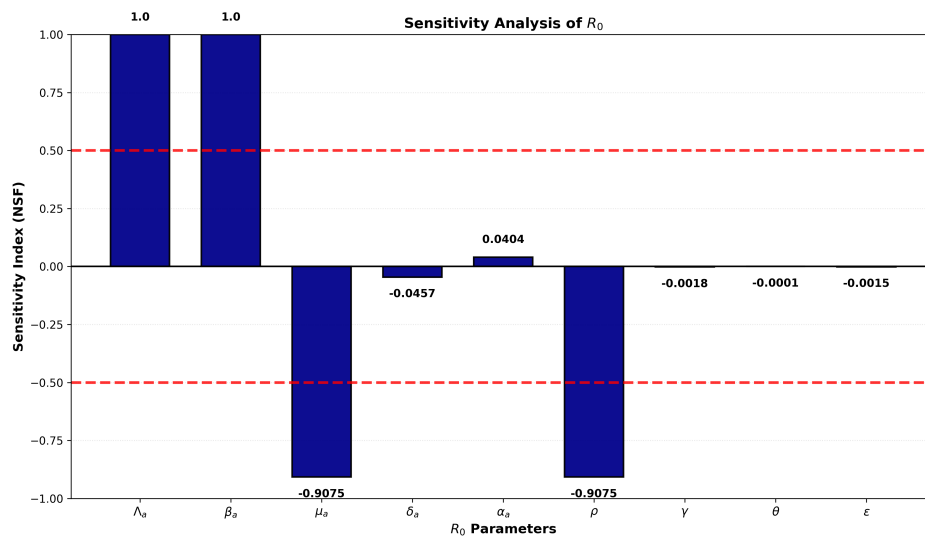


Figure 4.2: Sensitivity indices of parameters

Figure 4.2 shows how sensitive the different parameters are to $\mathcal{R}_0$. The height of the bars represents each parameter contribution to the basic reproduction number. It can be seen that recruitment rate (Λ_a) and the rate of infection of susceptible animals (β_a) are the most positive sensitive parameters while the most sensitive negative parameters are natural death and carcass removal rate (ρ).

Numerical Simulation. Numerical results were obtained using Python software to examine the dynamics of anthrax using parameters in table 4.2. Graphs are presented and then discussed below.

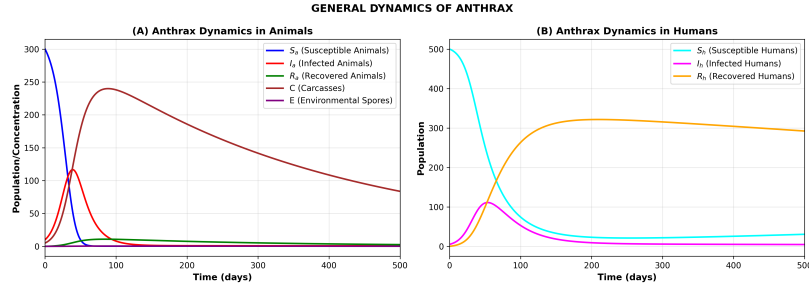


Figure 4.3: The general dynamics of anthrax in humans and animal populations.

As shown in figure 4.3, the population of Susceptible humans and animals reduces with time due to anthrax infection rate. Susceptible animals decrease as they come into contact with bacillus anthracis while grazing. The population of humans decrease as they acquire anthrax from the environment, eating meat from infected animals or by coming into contact with the carcass of an infected animal.

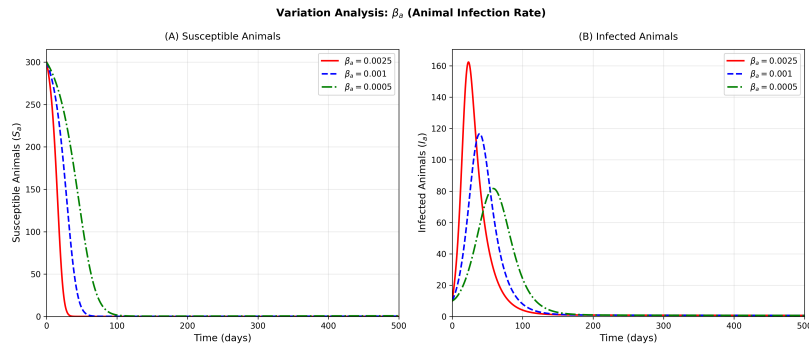


Figure 4.4: The dynamics of varying transmission rate (β_a) on animals population.

Figure 4.4 shows the impact of different animal transmission rates on the dynamics of anthrax in the animal population. It is observable that when animal infection rates rise, susceptible animals decline faster as compared to a low infection rate also the rise of infected animals is high when infection rates are high.

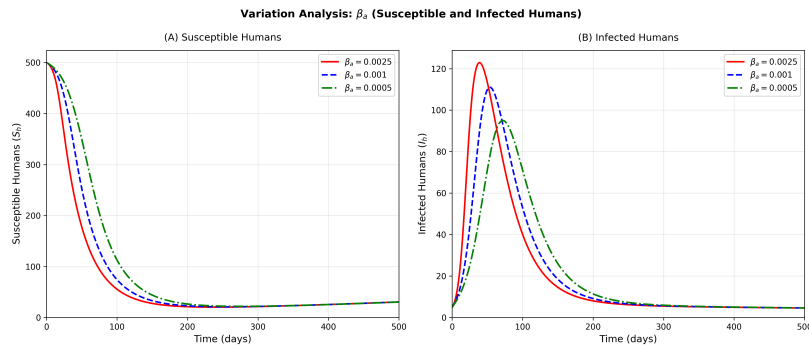


Figure 4.5: The dynamics of varying transmission rate (β_a) on human population.

As illustrated in Figure 4.4, figure 4.5, It is evident that the dynamics of the decline in susceptible humans differ from those of the animal population at the same rate of animal infection.

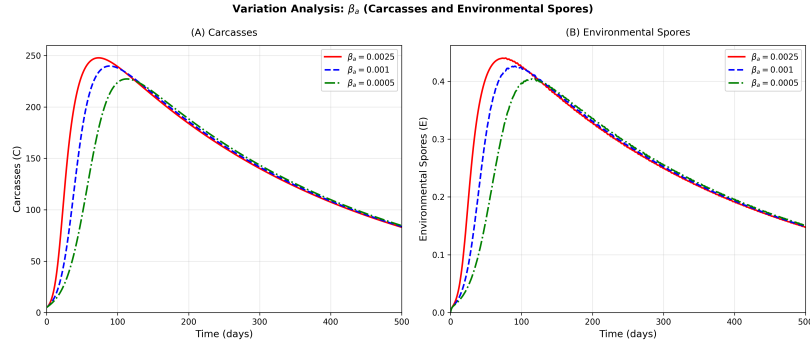


Figure 4.6: The dynamics of varying transmission rate (β_a) on carcass and environment.

Figure 4.6 illustrate the the effect of different animal transmission rate β_a on both carcass and the environment. The result show that a rise in β_a leads to a rise in carcass accumulation and environmental spores. This indicates that as the transmission rate among animals increases, more animals become infected, and as these infected animals die more carcasses are generated. Since carcasses are major sources of anthrax spores, their accumulation increases environmental spores.

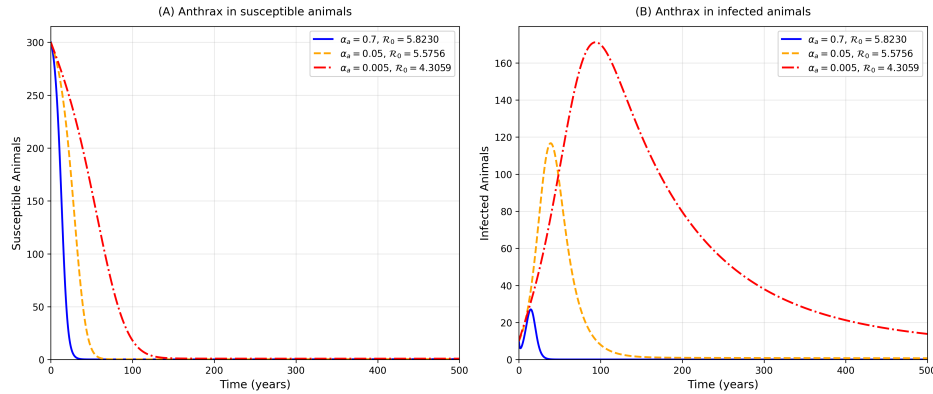


Figure 4.7: The dynamics of varying disease induced death rate (β_a) on animal population.

Figure 4.7 shows various trajectories of susceptible and infected animals for varying rates of disease induced death α_a . A higher rate of disease induced death rate α_a reduces peak of infection and then increases $\mathcal{R}_0$. This is because more carcasses are produced per infected animal, hence spores are then transmitted into the environment. This can be justified observing the graph at the left panel. The left panel shows that susceptible animals decrease at a lower rate when α_a is less because the associated $\mathcal{R}_0$ is also less but, when α_a is higher both susceptible animals decrease quick .

4.5 Optimal control

An optimal control problem is an optimization problem that, under initial conditions, minimizes or maximizes the objective functional subject to a system of equations [7]. In this study, we extended the model system (4.1.1) to included three control time dependent measures which are $u_1(t)$, $u_2(t)$, $u_3(t)$. $u_1(t)$ is the control measure aimed at increasing the immunity level of susceptible animals through vaccination. $u_2(t)$ is the control measure aimed at treatment of the infected animal population. This

helps to increase the rate of recovery δ_a . $u_3(t)$ is the safe carcass disposal control measure aimed at helping to reduce the spores produced from carcasses. All controls are bounded between 0 and 1:

$$0 \leq u_i(t) \leq 1, \quad i = 1, 2, 3, \quad \forall t \in [0, T].$$

By including the control measures into the system (4.1.1), we were able to achieve the optimal control model. By doing so we have the controlled system as:

$$\begin{cases} \frac{dS_a}{dt} = \Lambda_a - (1 - u_1)\beta_a(E + C)S_a - \mu_a S_a + \pi_a R_a, \\ \frac{dI_a}{dt} = (1 - u_1)\beta_a(E + C)S_a - (\delta_a + u_2 + \alpha_a + \mu_a)I_a, \\ \frac{dR_a}{dt} = (\delta_a + u_2)I_a - (\mu_a + \pi_a)R_a, \\ \frac{dS_h}{dt} = \Lambda_h - \beta_h(E + C + I_a)S_h - \mu_h S_h + \pi_h R_h, \\ \frac{dI_h}{dt} = \beta_h(E + C + I_a)S_h - (\delta_h + \mu_h + \alpha_h)I_h, \\ \frac{dR_h}{dt} = \delta_h I_h - (\mu_h + \pi_h)R_h, \\ \frac{dE}{dt} = (1 - u_3)(1 - \gamma)\rho C - (\theta + \varepsilon)E, \\ \frac{dC}{dt} = (\mu_a + \alpha_a)I_a - \rho(1 + u_3)C. \end{cases} \quad (4.5.1)$$

Objective Function. The objective function for the controlled system is defined as :

$$J(u_1, u_2, u_3) = \int_0^T \left[A_1 I_a(t) + A_2 I_h(t) + A_3 C(t) + \frac{B_1}{2} u_1^2(t) + \frac{B_2}{2} u_2^2(t) + \frac{B_3}{2} u_3^2(t) \right] dt, \quad (4.5.2)$$

where $A_1, A_2, A_3 > 0$ are weights constants associated with disease. $A_1 I_a(t), A_2 I_h(t), A_3 C(t)$ represent the cost of reducing infected animals, infected humans and carcass respectively. Also $B_i u_i^2$, $i = 1 \dots 3$ are the cost associated with these three control measures.

Optimal control characterization. Based on the optimal control equations (4.5.1) and (4.5.2) we then construct the Hamiltonian function for this study using the Pontryagin's principle. The Pontryagin's principle gives the conditions necessary for the controls to be optimal [7].

The Hamiltonian ($\mathcal{H}$). we have the Hamiltonian for this study as :

$$\left\{ \begin{array}{l} \mathcal{H} = A_1 I_a + A_2 I_h + A_3 C, \\ + \frac{B_1}{2} u_1^2 + \frac{B_2}{2} u_2^2 + \frac{B_3}{2} u_3^2, \\ + \lambda_1 [\Lambda_a - (1 - u_1) \beta_a (E + C) S_a - \mu_a S_a + \pi_a R_a], \\ + \lambda_2 [(1 - u_1) \beta_a (E + C) S_a - (\delta_a + u_2 + \alpha_a + \mu_a) I_a], \\ + \lambda_3 [(\delta_a + u_2) I_a - (\mu_a + \pi_a) R_a], \\ + \lambda_4 [\Lambda_h - \beta_h (E + C + I_a) S_h - \mu_h S_h + \pi_h R_h], \\ + \lambda_5 [\beta_h (E + C + I_a) S_h - (\delta_h + \mu_h + \alpha_h) I_h], \\ + \lambda_6 [\delta_h I_h - (\mu_h + \pi_h) R_h], \\ + \lambda_7 [(1 - u_3)(1 - \gamma) \rho C - (\theta + \varepsilon) E], \\ + \lambda_8 [(\mu_a + \alpha_a) I_a - \rho(1 + u_3) C]. \end{array} \right. \quad (4.5.3)$$

4.5.1 Theorem. Let u_1^*, u_2^*, u_3^* be optimal controls with corresponding optimal states $S_a^*, I_a^*, R_a^*, S_h^*, I_h^*, R_h^*, E^*, C^*$. Then there exist adjoint variables $\lambda_1, \lambda_2, \dots, \lambda_8$ satisfying the adjoint equations:

$$\dot{\lambda}_i = -\frac{\partial \mathcal{H}}{\partial x_i}, \quad \text{with transversal conditions } \lambda_i(T) = 0 \quad \text{for } i = 1, \dots, 8$$

from equation (4.5.3), we have the adjoint equations as:

$$\begin{aligned} \dot{\lambda}_1 &= -[-\lambda_1(\beta_a(1 - u_1)(E + C) + \mu_a) + \lambda_2\beta_a(1 - u_1)(E + C)], \\ \dot{\lambda}_2 &= -[A_1 - \lambda_2(\delta_a + u_2 + \alpha_a + \mu_a) + \lambda_3(\delta_a + u_2) + \lambda_8(\mu_a + \alpha_a) - \beta_h S_h(\lambda_5 - \lambda_4)], \\ \dot{\lambda}_3 &= -[\lambda_1\pi_a - \lambda_3(\mu_a + \pi_a)], \\ \dot{\lambda}_4 &= -[-\lambda_4(\beta_h(E + C + I_a) + \mu_h) + \lambda_5\beta_h(E + C + I_a)], \\ \dot{\lambda}_5 &= -[A_2 - \lambda_5(\delta_h + \mu_h + \alpha_h) + \lambda_6\delta_h], \\ \dot{\lambda}_6 &= -[\lambda_4\pi_h - \lambda_6(\mu_h + \pi_h)], \\ \dot{\lambda}_7 &= -[\beta_a(1 - u_1)S_a(\lambda_2 - \lambda_1) + \beta_h S_h(\lambda_5 - \lambda_4) - \lambda_7(\theta + \varepsilon)], \\ \dot{\lambda}_8 &= -[A_3 + \beta_a(1 - u_1)S_a(\lambda_2 - \lambda_1) + \beta_h S_h(\lambda_5 - \lambda_4) + (1 - u_3)(1 - \gamma)\rho\lambda_7 - \rho(1 + u_3)\lambda_8]. \end{aligned}$$

We then solve for the necessary conditions for optimizing the Hamiltonian. From equation (4.5.3) we obtain the optimal controls by solving $\frac{\partial \mathcal{H}}{\partial u_i} = 0, i = 1, \dots, 3$

Control u_1^* (Vaccination).

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial u_1} &= B_1 u_1^* + \lambda_1 \beta_a (E + C) S_a - \lambda_2 \beta_a (E + C) S_a \\ \Rightarrow \frac{\partial \mathcal{H}}{\partial u_1} &= B_1 u_1^* + \beta_a (E + C) S_a (\lambda_1 - \lambda_2) \\ u_1^* &= -\frac{\beta_a (E + C) S_a (\lambda_1 - \lambda_2)}{B_1} \\ u_1^*(t) &= \max \left\{ 0, \min \left\{ 1, \frac{\beta_a (E + C) S_a (\lambda_2 - \lambda_1)}{B_1} \right\} \right\}\end{aligned}$$

Control u_2^* (Treatment).

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial u_2} &= B_2 u_2 + I_a (\lambda_3 - \lambda_2) = 0 \\ \Rightarrow u_2^* &= \frac{I_a (\lambda_2 - \lambda_3)}{B_2}. \\ u_2^*(t) &= \max \left\{ 0, \min \left\{ 1, \frac{I_a (\lambda_2 - \lambda_3)}{B_2} \right\} \right\}.\end{aligned}$$

Control u_3^* (Carcass Disposal).

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial u_3} &= B_3 u_3 - (1 - \gamma) \rho C \lambda_7 - \rho C \lambda_8 = 0 \\ \Rightarrow u_3^* &= \frac{\rho C [\lambda_7 (1 - \gamma) + \lambda_8]}{B_3}. \\ u_3^*(t) &= \max \left\{ 0, \min \left\{ 1, \frac{\rho C [\lambda_7 (1 - \gamma) + \lambda_8]}{B_3} \right\} \right\}.\end{aligned}$$

Numerical simulation for optimal control. For simulation after incorporating control measures, this study employed the the forward-backward sweep algorithm introduced in Lenhart and Workman [7]. The numerical simulation was performed using the same parameters values presented in table (4.2). All possible combinations of the 3 control strategies u_1, u_2, u_3 , give a total of 8 optimal control scenarios. These 8 optimal control scenarios are grouped into 3 categories .

Category 1. This involves the numerical simulation of a single control scenarios. These scenarios are : Scenario 1 : $u_1 \neq 0, u_2 = 0, u_3 = 0$ Scenario 2 : $u_1 = 0, u_2 \neq 0, u_3 = 0$ Scenario 3 : $u_1 = 0, u_2 = 0, u_3 \neq 0$

Category 2. This category involves the simulation where two optimal control scenarios are not equal to zero . These scenarios are: Scenario 1 : $u_1 \neq 0, u_2 \neq 0, u_3 = 0$ Scenario 2 : $u_1 \neq 0, u_2 = 0, u_3 \neq 0$ Scenario 3 : $u_1 = 0, u_2 \neq 0, u_3 \neq 0$

Category 3. Refers to a single scenario, which is the simulation in which all optimal controls are put into place. In this case, $u_1 \neq 0, u_2 \neq 0, u_3 \neq 0$. This scenario is compared with the baseline. When there is no optimal control ($u_1 = u_2 = u_3 = 0$.)

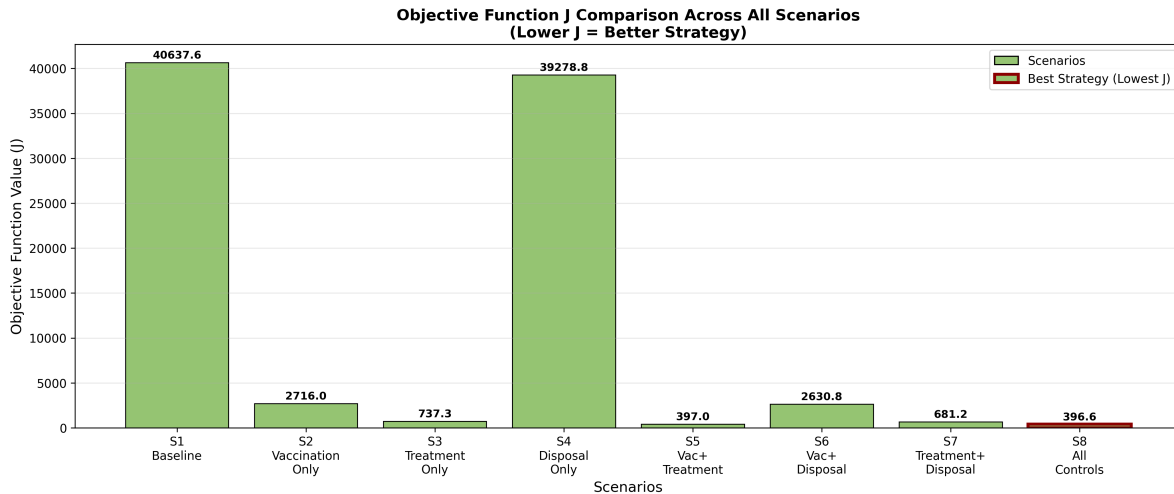


Figure 4.8: A chart showing the cost associated with the controls.

Figure 4.8 above presents the objective function (J) values across all 8 scenarios. The objective function J quantifies the total disease burden by combining disease costs and the costs of implementing all controls. A lower J value indicates that the control measure to be implemented is more effective. The baseline scenario (no controls) had $J = 40,637.6$. To identify the most effective control strategy, we compare each scenario to the baseline. This plot is a visualization of the total cost of combining disease burden and control implementation for each scenario. One graph from each category from the simulation is presented and discussed below.

Discussion of scenarios. After we show the overall cost for implementing each control measure, one graph from each category from the simulation is presented and discussed below.

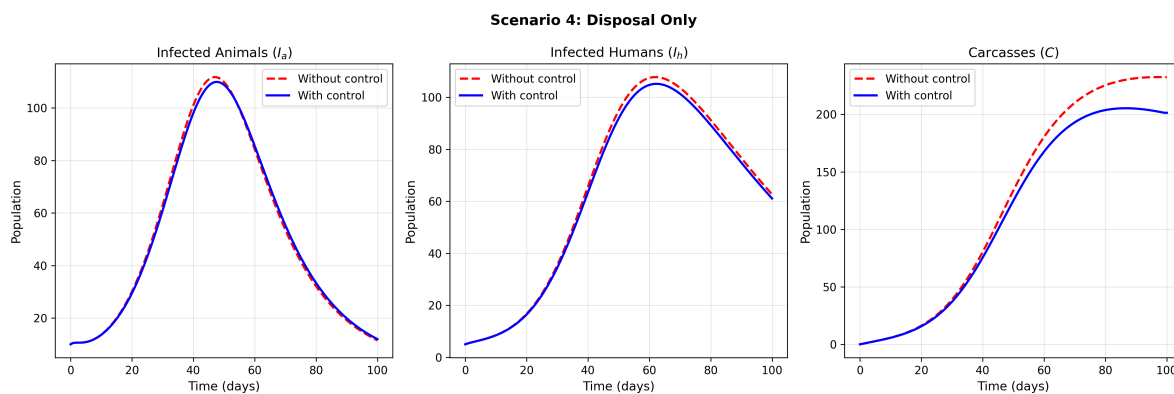


Figure 4.9: A simulation showing with and without carcass removal only.

Figure 4.9 shows the dynamics of implementing the control measure carcass disposal alone. It can be observed that this control measure provides a minimal impact. The infected animals I_a and infected humans I_h trajectories almost overlap the baseline dynamics, with only the carcass level C showing a modest reduction. As shown in Figure 4.8, implementing carcass disposal only achieved $J = 39,278.8$, representing a reduction of 1,358.8 (3.3%) from the baseline value $J = 40,637.6$.

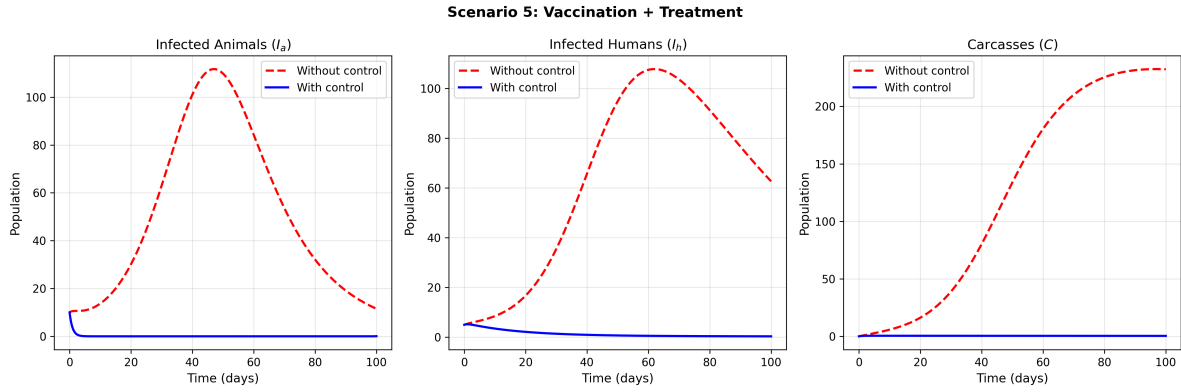


Figure 4.10: The simulation showing the dynamics of Vaccination and treatment.

Figure 4.10 shows the dynamics of implementing the combined control measures vaccination and treatment. As shown in Figure 4.8, implementing vaccination and treatment achieved $J = 397.0$, representing a reduction of 40,240.6 (99.0%) from the baseline value $J = 40,637.6$. This combination performs considerably better than vaccination alone ($J = 2,716.0$), showing that adding treatment to vaccination provides a substantial further benefit to the control of anthrax.

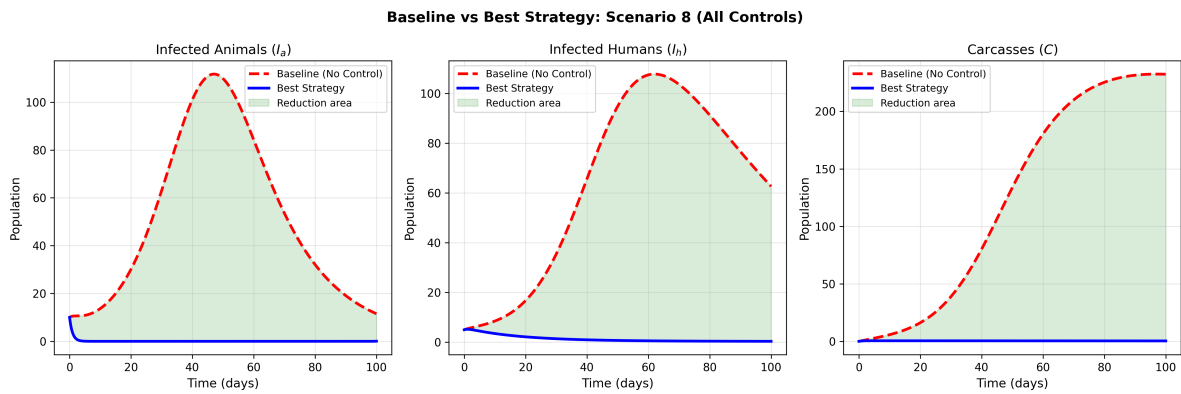


Figure 4.11: The simulation showing the dynamics of all the controls.

Figure 4.11 shows the baseline compared with the best strategy (all controls together). Implementing all three controls achieves the greatest decrease in disease burden. Infected humans I_h and carcass levels remain minimal throughout, and due to treatment, infected animals I_a also decrease and remain minimal. The shaded green areas represent the reduction in disease burden achieved by the controls. With this intervention, vaccination minimizes transmission, and treatment increases recovery (δ_a) while reducing disease-induced death (α_a), and disposal removes environmental reservoirs. As shown in Figure 4.8, Scenario 8 achieved $J = 396.6$, representing a reduction of 40,241.0 (99.0%) from the baseline value $J = 40,637.6$.

5. Conclusion and recommendation

5.1 Summary

In this work, an anthrax model was developed that captures animals, humans, the role of carcasses and environmental in anthrax transmission. The basic reproduction number ($\mathcal{R}_0$) of the formulated model was computed. The derived $\mathcal{R}_0$ expression (4.2.4) reveals that animals are the main drivers of anthrax transmission since no human parameters appear in the expression. This affirms the model assumption of no human to human transmission. The sensitivity index for each parameter in $\mathcal{R}_0$ was derived. The analysis shows that the rates of animal recruitment (Λ_a) and disease transmission (β_a) had the greatest positive impact on $\mathcal{R}_0$. This suggests that higher rates of animal recruitment (Λ_a) and disease transmission (β_a) increase the rate of new infections through carcass production and environmental transmission pathways. Based on sensitivity findings, this study proposes three control measures to eradicate anthrax. Vaccination(u_1) as a control measure to reduce β_a by protecting susceptible animals. Treatment(u_2) to help increase recovery rate of infected animals(δ_a) and reduce the disease-induced mortality rate(α_a). Carcass disposal(u_3) to increase ρ to reduce environmental transmission. The study's optimal control analysis shows that the best way to manage anthrax is to combine all three control interventions: vaccination (u_1), treatment (u_2), and carcass disposal (u_3).

Significance and Implications: This study provides a quantitative framework for understanding anthrax dynamics where livestock, wildlife, and humans coexist. Result of the studies identifies animal recruitment and transmission rates as key drivers, emphasizing reduced animal density, limited contact, carcass management, and environmental control. Overall, the study shows that anthrax transmission is fundamentally zoonotic, with environmental contamination playing a central role, especially in Africa where the disease remains a major public health and economic threat to livestock-dependent rural communities.

5.2 Recommendation

Based on the findings of this study, we recommend policy makers and public health stakeholders to raise public awareness of the need for veterinary services. To be able to eliminate anthrax, we suggest doing the following particular actions. Educate and encourage farmers on the need to vaccinate animals. Secondly, it is important to promote veterinary visits at regular intervals to enable early diagnosis for treatment of infected animals to help limit transmission. Also, we recommend that carcasses should be disposed of properly and environmental decontamination, such as fumigation of anthrax endemic areas, to help prevent environmental spore accumulation. Finally, raising public awareness about anthrax symptoms and safe handling practices can help reduce risky behaviors in communities.

5.3 Future work

Our future work ought to:

- study the transmission of anthrax using a fractional approach.
- Incorporate incubation periods to capture early dynamics.

Appendix A. Some additional computations

Theorem (4.1.2). We proof the boundedness of the human population. We consider the human Population of the system (4.1.1). From (4.1.1) we have: $\square$

$$\begin{aligned} N_h(t) &= S_h(t) + I_h(t) + R_h(t) \\ \frac{dN_h(t)}{dt} &= \frac{dS_h(t)}{dt} + \frac{dI_h(t)}{dt} + \frac{dR_h(t)}{dt} \end{aligned}$$

$$\implies \frac{dN_h}{dt} = \Lambda_h - \beta_h(E+C)S_h - \mu_h S_h + \pi_h R_h + \beta_h(E+C)S_h - (\delta_h + \alpha_h + \mu_h)I_h + \delta_h I_h - (\mu_h + \pi_h)R_h$$

$$\begin{aligned} \implies \frac{dN_h}{dt} &= \Lambda_h - \mu_h(S_h + I_h + R_h) - \alpha_h I_h \\ \implies \frac{dN_h}{dt} &= \Lambda_h - \mu_h(N_h(t)) - \alpha_h I_h \\ \implies \frac{dN_h}{dt} &\leq \Lambda_h - \mu_h(N_h(t)), \quad \text{since } I_h \geq 0 \\ \implies \frac{dN_h}{dt} + \mu_h(N_h(t)) &\leq \Lambda_h \end{aligned}$$

let $e^{\mu_h t}$ be an integration factor. and then we multiply through by the integration factor.

$$\begin{aligned} e^{\mu_h t} \frac{dN_h}{dt} + \mu_h(N_h(t))e^{\mu_h t} &\leq \Lambda_h e^{\mu_h t} \\ \implies \int_0^t \frac{d}{ds} (N_h(s)e^{\mu_h s}) ds &\leq \int_0^t \Lambda_h e^{\mu_h s} ds \\ \implies [N_h(s)e^{\mu_h s}]_0^t &\leq \left[\Lambda_h \frac{1}{\mu_h} e^{\mu_h s} \right]_0^t \\ \implies N_h(t)e^{\mu_h t} - N_h(0) &\leq \frac{\Lambda_h}{\mu_h} [e^{\mu_h t} - 1] \\ \implies N_h(t) &\leq \frac{\Lambda_h}{\mu_h} - \frac{\Lambda_h}{\mu_h} e^{-\mu_h t} + N_h(0)e^{-\mu_h t} \end{aligned}$$

$N_h(t) \geq 0$ because for all $t \geq 0$ $S_h(t), I_h(t), R_h(t) \geq 0$ and also

$$\begin{aligned} N_h(t) &= S_h(t) + I_h(t) + R_h(t) \\ \implies 0 \leq N_h(t) &\leq \frac{\Lambda_h}{\mu_h} + e^{-\mu_h t} \left[N_h(0) - \frac{\Lambda_h}{\mu_h} \right] \end{aligned}$$

As $t \rightarrow +\infty$, we have:

$$\begin{aligned} 0 &\leq \lim_{t \rightarrow \infty} N_h(t) \leq \lim_{t \rightarrow \infty} \left(\frac{\Lambda_h}{\mu_h} + e^{-\mu_h \cdot t} \left[N_h(0) - \frac{\Lambda_h}{\mu_h} \right] \right) \\ &\implies 0 \leq N_h(t) \leq \frac{\Lambda_h}{\mu_h} \quad \forall t \geq 0 \end{aligned}$$

Theorem (4.1.1). We proof the positivity of other compartments not in the main work. $\square$

We consider the infected animal compartment. The infected animal compartment from (4.1.1) is given by:

$$\begin{aligned} \frac{dI_a}{dt} &= \beta_a(E + C)S_a - (\delta_a + \alpha_a + \mu_a)I_a. \\ \implies \dot{I}_a(t) &\geq -(\delta_a + \alpha_a)I_a \quad \text{for} \\ \implies \frac{\dot{I}_a(t)}{I_a} &\geq -(\delta_a + \alpha_a), \quad I_a \neq 0. \\ \implies \int \frac{\dot{I}_a(t)}{I_a} dt &\geq - \int (\delta_a + \alpha_a) dt. \\ \implies \ln I_a(t) &\geq -(\delta_a + \alpha_a)t + K \\ \implies I_a(t) &\geq e^K \cdot e^{-(\delta_a + \alpha_a)t}. \end{aligned}$$

Evaluating at $t = 0$, $e^K = I_a(0)$:

$$I_a(t) \geq I_a(0) e^{-(\delta_a + \alpha_a + \mu_a)t} \geq 0.$$

Since $I_a(0) \geq 0$ and the exponential term is always positive, $I_a(t) \geq 0$ for all $t \geq 0$.

We consider the recovered Animal Compartment. The equation for the recovered animals compartment from (4.1.1) is given by:

$$\begin{aligned} \frac{dR_a}{dt} &= \delta_a I_a - (\mu_a + \pi_a)R_a \\ \implies \dot{R}_a(t) &\geq -(\mu_a + \pi_a)R_a(t) \\ \implies \frac{\dot{R}_a(t)}{R_a(t)} &\geq -(\mu_a + \pi_a), \quad R_a(t) \neq 0 \\ \implies \int \frac{\dot{R}_a(t)}{R_a(t)} dt &\geq - \int (\mu_a + \pi_a) dt \\ \implies \ln R_a(t) &\geq -(\mu_a + \pi_a)t + K \\ \implies R_a(t) &\geq e^K \cdot e^{-(\mu_a + \pi_a)t} \end{aligned}$$

Evaluating at $t = 0$, $e^K = R_a(0)$:

$$R_a(t) \geq R_a(0) e^{-(\mu_a + \pi_a)t} \geq 0$$

Since $R_a(0) \geq 0$ and the exponential term is always positive, $R_a(t) \geq 0$ for all $t \geq 0$.

We consider the Susceptible human Compartment. From (4.1.1) we have:

$$\frac{dS_h}{dt} = \Lambda_h - \beta_h(E + C + I_a)S_h - \mu_h S_h + \pi_h R_h$$

Since $\Lambda_h \geq 0$ and $\pi_h R_h \geq 0$:

$$\begin{aligned} \implies \dot{S}_h(t) &\geq -(\beta_h(E + C + I_a) + \mu_h)S_h(t) \\ \implies \frac{\dot{S}_h(t)}{S_h(t)} &\geq -(\beta_h(E + C + I_a) + \mu_h), \quad S_h(t) \neq 0 \\ \implies \int \frac{\dot{S}_h(t)}{S_h(t)} dt &\geq - \int (\beta_h(E + C + I_a) + \mu_h) dt \\ \implies \ln S_h(t) &\geq - \int (\beta_h(E + C + I_a) + \mu_h) dt + \ln K \\ \implies S_h(t) &\geq K \cdot e^{-\int (\beta_h(E+C+I_a)+\mu_h) dt} \end{aligned}$$

Evaluating at $t = 0$, $K = S_h(0)$:

$$S_h(t) \geq S_h(0) e^{-\int_0^t (\beta_h(E+C+I_a)+\mu_h) dt} > 0$$

Since $S_h(0) > 0$ and the exponential term is always positive, $S_h(t) > 0$ for all $t \geq 0$.

We consider the Infected human Compartment. From (4.1.1) we have:

$$\frac{dI_h}{dt} = \beta_h(E + C + I_a)S_h - (\delta_h + \mu_h + \alpha_h)I_h$$

$$\begin{aligned} \implies \dot{I}_h(t) &\geq -(\delta_h + \mu_h + \alpha_h)I_h(t) \\ \implies \frac{\dot{I}_h(t)}{I_h(t)} &\geq -(\delta_h + \mu_h + \alpha_h), \quad I_h(t) \neq 0 \\ \implies \int \frac{\dot{I}_h(t)}{I_h(t)} dt &\geq - \int (\delta_h + \mu_h + \alpha_h) dt \\ \implies \ln I_h(t) &\geq -(\delta_h + \mu_h + \alpha_h)t + K \\ \implies I_h(t) &\geq e^K \cdot e^{-(\delta_h+\mu_h+\alpha_h)t} \end{aligned}$$

Evaluating at $t = 0$, $e^K = I_h(0)$:

$$I_h(t) \geq I_h(0) e^{-(\delta_h+\mu_h+\alpha_h)t} \geq 0$$

Since $I_h(0) \geq 0$ and the exponential term is always positive, $I_h(t) \geq 0$ for all $t \geq 0$.

We consider the recovered human Compartment. From (4.1.1) we have:

$$\frac{dR_h}{dt} = \delta_h I_h - (\mu_h + \pi_h)R_h$$

$$\begin{aligned}
&\implies \dot{R}_h(t) \geq -(\mu_h + \pi_h)R_h(t) \\
&\implies \frac{\dot{R}_h(t)}{R_h(t)} \geq -(\mu_h + \pi_h), \quad R_h(t) \neq 0 \\
&\implies \int \frac{\dot{R}_h(t)}{R_h(t)} dt \geq - \int (\mu_h + \pi_h) dt \\
&\implies \ln R_h(t) \geq -(\mu_h + \pi_h)t + K \\
&\implies R_h(t) \geq e^K \cdot e^{-(\mu_h + \pi_h)t}
\end{aligned}$$

Evaluating at $t = 0$, $e^K = R_h(0)$:

$$R_h(t) \geq R_h(0) e^{-(\mu_h + \pi_h)t} \geq 0$$

Since $R_h(0) \geq 0$ and the exponential term is always positive, $R_h(t) \geq 0$ for all $t \geq 0$.

We use Integration factor method to proof the positivity of the environmental compartment. From (4.1.1) we have the equation for the environmental compartment as:

$$\frac{dE}{dt} = (1 - \gamma)\rho C - (\theta + \varepsilon)E$$

Since $(1 - \gamma)\rho C \geq 0$:

$$\begin{aligned}
&\implies \frac{dE}{dt} \geq -(\theta + \varepsilon)E(t) \\
&\implies \frac{dE}{dt} + (\theta + \varepsilon)E(t) \geq 0
\end{aligned}$$

Let $e^{(\theta + \varepsilon)t}$ be an integration factor. Multiply both sides by $e^{(\theta + \varepsilon)t}$:

$$\begin{aligned}
&\implies e^{(\theta + \varepsilon)t} \frac{dE}{dt} + (\theta + \varepsilon)E(t)e^{(\theta + \varepsilon)t} \geq 0 \\
&\implies \frac{d}{dt} \left(E(t)e^{(\theta + \varepsilon)t} \right) \geq 0 \\
&\implies \int_0^t \frac{d}{ds} \left(E(s)e^{(\theta + \varepsilon)s} \right) ds \geq 0 \\
&\implies \left[E(s)e^{(\theta + \varepsilon)s} \right]_0^t \geq 0 \\
&\implies E(t)e^{(\theta + \varepsilon)t} - E(0)e^0 \geq 0 \\
&\implies E(t)e^{(\theta + \varepsilon)t} \geq E(0)
\end{aligned}$$

Therefore:

$$E(t) \geq E(0)e^{-(\theta + \varepsilon)t} \geq 0$$

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